

The Human Capital Effects of Housing Assistance*

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Abstract

This paper estimates the impact of means-tested housing subsidies on children’s human capital accumulation and early adult outcomes. I link applications to public housing and housing voucher wait lists in New Haven, Connecticut, to statewide education, labor market, and criminal justice records, and exploit variation in when households reach the top of randomly-ordered wait lists as an instrument for childhood exposure to assistance. Receiving housing assistance nearly eliminates school-reported homelessness and housing instability at younger ages and raises measures of early childhood development among boys. Housing assistance also improves longer-run outcomes. Each additional year of assistance raises high school graduation and college enrollment by 2 to 3 percentage points, and children that receive housing assistance are less likely to be convicted of a felony in early adulthood. To distinguish among mechanisms driving these results, I estimate effects on school continuity, school quality, and adult labor supply. Finally, translating the attainment effects into predicted impacts on lifetime earnings and accounting for reductions in incarceration and sheltered homelessness, I calculate that providing housing assistance to a household with children produces roughly \$2 in benefits per government dollar spent. Because of larger impacts on educational attainment, housing vouchers are more cost-effective than public housing units.

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1 Introduction

Almost 4 million children in the United States live in households receiving means-tested federal housing subsidies. These transfers are large and long-lasting: the average household receives more than \$13,600 annually and nearly two-thirds of non-elderly recipients keep the subsidies for at least a decade (Husock and Meyer 2025). Although assistance can span much of childhood, whether it improves children’s long-run outcomes – and does so cost-effectively – remains unclear.

Housing assistance may raise children’s human capital by preventing homelessness and residential instability, enabling moves to higher-opportunity neighborhoods, or improving access to higher-quality schools.¹ In practice, gains are limited by the design of housing assistance and by frictions in the housing market. Long wait lists delay assistance (Collinson et al. 2015; Waldinger 2021), while search frictions and jurisdictional boundaries limit voucher recipients’ access to high-opportunity neighborhoods (Bergman et al. 2024; Garboden 2024). Poor unit conditions in public housing may also harm children, as evidenced by longstanding concerns among policymakers that public housing is “bad for kids” (Currie and Yelowitz 2000; Collinson and Gibbs 2025). Finally, subsidies may reduce parental labor supply and earnings (Jacob and Ludwig 2012; van Dijk 2019), with potentially offsetting effects on the time and money parents invest in children.

In this paper, I combine new data with a natural experiment to provide new evidence on the effects of housing vouchers and public housing on children, focusing on their human capital accumulation, educational attainment, and early adult criminal justice outcomes. First, I construct a new administrative panel linking more than a decade of housing assistance wait list applications in New Haven, Connecticut, to statewide education, labor market, and criminal justice records. Second, I develop a research design based on a series of lotteries that randomly ordered applicants on wait lists for housing vouchers and public housing units. I use lottery-induced variation in the date a household is selected from these wait lists as an instrument for whether a child receives assistance and, for longer-run outcomes, the number of childhood years spent with housing subsidies.

My primary analysis relies on application records for more than 5,500 children living in New Haven county linked to administrative data on annual school-reported homelessness and attendance, kindergarten readiness, standardized achievement tests in elementary and middle school, high school graduation, and college enrollment. These linked

¹The neighborhood channel may be particularly important, given evidence that childhood exposure to neighborhoods causally impacts long-run outcomes (Chetty et al. 2016; Chyn 2018; Katz 2026).

records allow me to evaluate the effects of housing assistance on residential stability, human capital accumulation, and educational attainment. Because lotteries occur over time and children differ in age at the time of application, I use different subsamples across outcomes. I additionally link adult household members to quarterly earnings records to characterize changes in household resources, and children to felony conviction records to examine early adult criminal justice outcomes.²

I estimate the causal impacts of housing vouchers and public housing on children using nine wait list position lotteries held between 2003 and 2018. Because randomly-assigned positions were often overwritten, I instrument for receiving assistance with the timing of selection for an eligibility interview, which I refer to as an “offer”.³ In separate estimates for children from ages six to seventeen, an offer raises annual assistance receipt by more than 45 percentage points at any age, with first-stage *F*-statistics close to 1,000. Because the timing of offers depends on an applicant’s randomly-assigned position and on unit or voucher turnover, I check for balance with respect to household characteristics that may also affect child outcomes. After including fixed effects that absorb differences in offer timing across lotteries for households with differing unit eligibility, I find no remaining differences in pre-application demographic and labor market characteristics between children in offered and not-offered households at any age.

Using this research design, I first show that housing assistance substantially reduces children’s exposure to school-reported homelessness and housing instability. Pooling annual observations from ages six through eleven, receiving assistance by a given age reduces the probability of school-reported homelessness or housing instability by 4.4 percentage points ($p = 0.002$), relative to a complier control mean of 6.6%. The reduction is largest at ages six to seven, when assistance lowers this probability by 7.3 percentage points ($p < 0.001$), relative to a complier control mean of 9.2%. Estimated reductions remain negative at older ages, although precision declines with age. Pooling ages six through eleven, assistance reduces non-shelter housing instability by 2.9 percentage points ($p = 0.016$), driven primarily by a 2.9 percentage point reduction in doubled-up homelessness ($p = 0.008$). The estimated reduction in sheltered homelessness is 1.5 percentage points and is marginally significant ($p = 0.055$), while effects on living in a hotel or motel – a rare outcome – are small and statistically insignificant. These findings complement [Collinson and Gibbs 2025](#), who find that both housing vouchers and pub-

²Because of my Data Use Agreement (DUA) with DataLinkCT, I am unable to directly combine the criminal justice records with the linked education and labor market data.

³Roughly 60% of children in the sample are in households that eventually receive an offer before they reach the age of 18. Binary treatment specifications use an offer indicator, while exposure specifications use years since selection for the orientation interview.

lic housing reduce sheltered homelessness and doubling up among families with young children in Miami-Dade county.

Second, using records for more than 2,000 children that enter kindergarten after a wait list lottery, I find evidence that housing assistance improves boys' early developmental outcomes. Boys whose families receive assistance before kindergarten score 0.21 standard deviations higher on an index averaging teachers' assessments of kindergarten readiness across six domains ($p = 0.113$). The largest point estimates are for creativity, physical and motor development, and personal and social development – the effect on an average of these skills is 0.31 standard deviations ($p = 0.056$). A complementary latent factor model that separates these developmental skills from language, literacy, and numeracy yields a 0.35 standard deviation improvement for boys ($p = 0.040$). In contrast, the estimated effect on academic skills is small and statistically insignificant, and effects for girls are imprecisely estimated. If anything, public housing units appear to have larger effects than voucher on developmental skills ($p = 0.062$). The concentration of developmental gains among boys parallels [Collinson and Gibbs 2025](#), who find that the positive effects of housing vouchers on early literacy and kindergarten readiness are driven primarily by boys. Despite this, I find no consistent effects on elementary or middle school achievement for either gender. This pattern resembles evidence of test score “fade out” from studies of Head Start ([Currie and Thomas 1995](#); [Deming 2009](#)).

Next, I use a partially overlapping sample of roughly 2,000 children that reach ages twenty to twenty-two by 2024 to examine the longer-run effects of housing assistance, focusing on educational attainment and criminal justice involvement in early adulthood. Among compliers, housing assistance increases high school graduation by age 20 by 14.9 percentage points ($p = 0.145$) and college enrollment by age 20 by 19.1 percentage points ($p = 0.062$). To exploit the amount of variation in childhood exposure to assistance, I estimate a dose-response design and find that, under a constant marginal effects assumption ([Chyn 2018](#); [Chaudhry and Eng 2024](#)), each additional year of assistance increases high school graduation by age 22 by 2.9 percentage points ($p = 0.099$) and college enrollment by age 22 by 4.2 percentage points ($p = 0.020$). The effect on college enrollment is concentrated in the four-year sector. These attainment gains are concentrated among voucher recipients, and I find suggestive evidence of larger college enrollment effects among girls. Finally, housing assistance reduces the probability of a felony conviction by age 22 by 7.4 to 9.8 percentage points ($p = 0.097$, $p = 0.045$), with the effects concentrated among boys. Together, these results indicate that housing assistance can generate lasting gains in educational attainment and reduce involvement with the criminal justice system.

To better understand these longer-run effects, I examine mechanisms related to school attachment, school quality, and household resources. Housing assistance strengthens children’s attachment to school early in childhood, raising attendance and reducing chronic absenteeism from ages six to eleven. These improvements do not appear to operate through access to higher-performing schools, and I find small and imprecise effects on school test score value added or average proficiency in elementary and middle school. If anything, assisted children attend schools with larger shares of economically disadvantaged students at older ages. For example, assistance increases the share of peers with free or reduced-price lunch share by 6 to 7 percentage points from ages fourteen to seventeen. In sum, the attainment gains appear more consistent with increased housing stability and school continuity than with movement to observably better schools. Effects on adult earnings are likewise small and insignificant, both along the intensive and extensive margin.

Finally, to benchmark the cost-effectiveness of government spending on housing assistance, I map the estimated effects on educational attainment into predicted impacts on lifetime earnings using estimates from [Zimmerman 2014](#) and [Hendren and Sprung-Keyser 2020](#), and I use other external estimates to calibrate the reductions in government costs from sheltered homelessness and crime ([McCollister et al. 2010](#)). I find that housing assistance returns between \$2.05 and \$2.26 per dollar of net government spending, depending on whether assistance is valued at \$0.83 per dollar or equivalently to cash ([Reeder 1985](#)). Under a baseline assumption that vouchers and units are valued by households equivalently to cash, I estimate that each dollar spent on housing vouchers returns around \$2 in benefits, while each dollar spent on public housing returns around \$1. In my setting, differences in the MVPF arise from larger effects on college enrollment among voucher recipients.

Related literature This paper contributes to two strands of research. First, it adds to a large literature that studies how means-tested housing assistance affects children. Researchers often exploit variation in childhood exposure to public housing or housing vouchers and find positive effects on outcomes ranging from test scores to early-adult earnings and health outcomes ([Currie and Yelowitz 2000](#); [Schwartz et al. 2020](#); [Pollakowski et al. 2022](#); [Chaudhry and Eng 2024](#); [Han and Schwartz 2024](#); [Beauregard 2025](#); [Ribeiro and Mariante 2025](#)). A smaller set of studies uses lottery-ordered wait lists to estimate the effects of housing assistance on adults ([Jacob and Ludwig 2012](#); [van Dijk 2019](#); [Carr and Koppa 2020](#)) and children ([Jacob et al. 2013, 2015](#)), finding mixed results. The closest paper to this one is [Collinson and Gibbs 2025](#), which studies a combined housing voucher and public housing lottery in Miami-Dade County and finds improvements in early child-

hood skills, concentrated among families that move to higher-opportunity neighborhoods or higher-quality public housing developments. I add to this literature by combining data on longer-run schooling and criminal justice outcomes, allowing me to examine changes in housing stability, school attachment, educational attainment, criminal justice involvement, and household resources within the same setting.

Second, this paper contributes to a broad literature on in-kind transfers and children's long-run outcomes, especially their human capital development. Most transfers provided to low-income families in the United States are in kind rather than in cash (Currie and Gahvari 2008), and a large empirical literature finds that childhood access to the social safety net improves later-life health, educational attainment, and earnings (Currie and Thomas 1995; Kline and Walters 2016; Hoynes et al. 2016; Brown et al. 2020; Bailey et al. 2021; Gray-Lobe et al. 2023; Bailey et al. 2024; Currie and Van Parys 2026). My results suggest that protection against adverse shocks, such as homelessness, may be one channel through which in-kind transfers improve long-run outcomes, even in the absence of significant gains in school-age test scores.

The remainder of the paper proceeds as follows: Section 2 describes the institutional context. Section 3 describes the data sources and samples. Section 4 introduces the research design. Section 5 presents the main results and examines possible mechanisms. Section 7 discusses the cost effectiveness of housing assistance. Section 8 concludes.

2 Background

2.1 Housing assistance programs and Elm City Communities

The Department of Housing and Urban Development (HUD) provides long-term rental assistance for low-income households in two categories: rental vouchers, known as *Housing Choice Vouchers* (HCVs), and subsidies tied to specific units. HCVs subsidize leases of qualifying units in the private rental market. Unit-based assistance includes traditional *Low Income Public Housing* (LIPH) and *Project Based Vouchers* (PBVs), both of which subsidize leases in specific developments. These programs are funded by HUD but administered by more than 3,300 local Public Housing Agencies (PHAs). In 2023, federal housing assistance subsidized around 4.5 million households at an annual cost of more than \$54 billion.⁴

⁴National and local program statistics in this subsection are calculated from the 2023 edition of HUD's [Picture of Subsidized Households](#) dataset.

These two forms of assistance share similar eligibility and subsidy rules but differ in how they provide housing subsidies.⁵ In both programs, eligibility depends primarily on household income, and recipients contribute 30% of adjusted household income toward rent.⁶ For tenant-based vouchers, the PHA pays a participating landlord the difference between the tenant's contribution and the unit's contract rent, up to a locally determined payment standard, typically set at an estimate of the 40th percentile rents in a metropolitan area. In order to successfully use a voucher, recipients must locate a qualifying unit and a landlord willing to accept the voucher.⁷ In the LIPH and PBV programs, the subsidy is attached to an apartment either owned by the PHA or operated through a long term agreement between the PHA and the property owner. Recipients of unit-based assistance face lower search frictions in the rental housing market but have less choice over the unit and neighborhood in which they live (Ellen et al. 2024).

This study examines housing assistance administered by Elm City Communities (ECC), the PHA serving New Haven, Connecticut. New Haven is a relatively low-income city surrounded by a more affluent metropolitan area: in 2023, median household income in New Haven was roughly \$51,000, compared with almost \$81,000 in the surrounding area, and around one-in-four residents lived below the poverty line. ECC provided assistance to nearly 13,500 residents in 2023, around 10% of the city's estimated population of 135,319.

Housing assistance provided by ECC is a large transfer relative to recipients' resources. In 2023, the average ECC household received an annual subsidy of \$14,292 and paid \$5,520 toward rent. Average household income, including cash transfers and other non-wage income, was only \$19,234, far below even the income eligibility cutoff for one-person households of \$40,150. This pattern is consistent with nationwide evidence that housing assistance recipients are negatively-selected on income relative to the eligible population (Rafkin et al. 2025).

⁵Households are eligible for HCVs when their annual income is below 50% of the median income in their metropolitan area (AMI). Households are eligible for PBVs and LIPH when their annual income is below 80% of AMI.

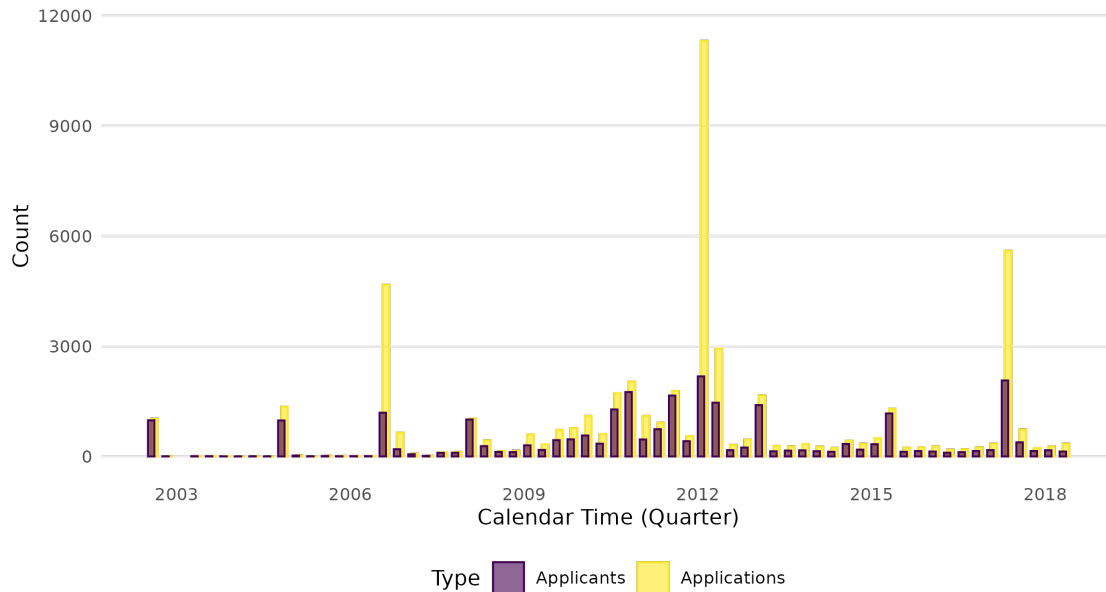
⁶Household income is adjusted for children, elderly or disabled household members, and medical hardships. Households may deduct \$480 per dependent and \$400 per elderly or disabled household member from their reported annual income. Households may deduct medical expenses whenever they exceed 3% of annual income.

⁷A unit will be declared ineligible if it either (i) fails a health and safety inspection or (ii) is determined to not have a "reasonable" contract rent (i.e., one in line with market rents for comparable rents).

2.2 ECC wait lists

Unlike many social insurance programs in the United States, housing assistance is not an entitlement: income-eligible households are not guaranteed a voucher or unit, and local PHAs use wait lists to ration assistance. Wait list policies vary across PHAs, from indefinitely open, first-in-first-out queues to lists that accept applications only during infrequent opening periods (Collinson et al. 2015; Waldinger 2021). During the 2003 to 2018 period that I study, ECC typically opened wait lists only when it either expected that the existing queue could be processed within a few months (Bromage 2005) or needed to fill units in recently remodeled developments. ECC announced the beginning and end of each application period through newspapers, flyers, and online communications. With limited exceptions, applicants could join a list only while it remained open. Appendix B describes these opening periods in more detail.

Figure 1: ECC wait list applications over time



Note: The y-axis is the count of applications (applicants) to one of the 24 primary Elm City Communities wait lists for non-elderly households. The x-axis is the calendar time in quarters from January 2003 to December 2018. The data source is universe of ECC wait list applications.

ECC used two methods to order applicants during the study period. During some openings, applicants were ordered by the exact date and time of application (i.e., first-in-first-out). During others, ECC assigned each applicant a random number that determined whether they were placed on the wait list and, conditional on placement, their initial wait list position. Applicants assigned numbers below a predetermined cutoff were added to

the list in random number order behind any existing applicants. ECC does not retain records of unplaced applicants.

ECC uses a combination of site-specific and general wait lists to allocate vouchers and unit-based assistance. From 2003 to 2018, ECC received more than 75,000 applications from more than 25,000 households across 81 wait lists. As shown in Figure 1, the number of applications noticeably spikes during, for example, a 2005 housing voucher wait list opening and a set of 2007 public housing wait list openings. Across the 24 primary wait lists for non-elderly households, the average number of applications per wait list is 2,160, or 34 per quarter.

3 Data and matching

The data in this paper combine wait list application and income certification records from ECC with administrative data accessed through the Connecticut Office of Policy and Management’s DataLinkCT initiative and public records from the Connecticut Judicial Branch. I describe these data sources below.

Housing assistance data I obtain data on applications to housing assistance wait lists from Elm City Communities (ECC). ECC is the only Public Housing Agency in New Haven, Connecticut. For each applicant-wait list combination, I observe the exact date and time of application, the name of the wait list, the household’s wait list position after application, and the household’s most recent reported address. Each application record contains information on every household member’s gender, age, and race.⁸ For applicants that receive housing assistance from ECC, I additionally observe annual income certification records that are used to verify household eligibility and calculate rent payments.⁹ Certification records also contain information on point-in-time household address and formal and informal earnings. Finally, I observe dates of participation in various community and economic development (CED) programs run by ECC.

Education data I obtain data on school attendance and test scores from the Connecticut State Department of Education (SDE), which maintains records on public K-12 educational programs and postsecondary enrollment for Connecticut public high school graduates. DataLinkCT matches the children of applicants to ECC wait lists to SDE records on the universe of New Haven county public school children from the 2010-2011 to 2023-

⁸Each application contains the date of birth for all applicant household members. My Data Use Agreement (DUA) with DataLinkCT does not allow me to observe individual ages in the matched sample.

⁹These records form the basis for the Form 50058 submissions made to HUD.

2024 school years using name, date of birth, and the last 4 digits of the Social Security number (SSN).¹⁰ Importantly, a student is observed in the matched file in each year they attend *any* Connecticut public school if they *ever* attend a public school within New Haven county. SDE records include information on kindergarten achievement tests, end-of-year test scores for students in grades 3-8, annual measures of students' school attendance and housing insecurity, and the date of high school graduation. Kindergarten test scores, unlike the remaining education records, are available as early as the 2008-2009 school year. Test scores in grades 3-8 are only available beginning in the 2014-2015 school year.

In addition to the K-12 education records, I observe information on enrollment in postsecondary education for the universe of Connecticut public high school graduates that ever attend a New Haven county school. These students are matched to the National Student Clearinghouse (NSC) if they have postsecondary enrollment dates from May 2011 to August 2024. The NSC file contains information on enrollment dates, enrollment status, and institution level. For students who successfully graduate, I observe the degree type, major, and graduation date. The data cover both two-year and four-year degree programs. I describe the SDE data files in more detail in Appendix C.1.

Labor market data I also obtain data on quarterly earnings and employment from 2001-2021 from the Connecticut Department of Labor (DOL), which maintains records of earnings reported by employers in Connecticut. DataLinkCT matches adult household members in ECC wait list applications to DOL records on the universe of Connecticut residents. I keep only unique matches and inflate earnings to be in 2023 dollars. Employer identifiers are not available, but the data does record the employment sector (i.e., 6-digit NAICS code). I describe the DOL data file in more detail in Appendix C.2. I use these records to measure pre-application household employment and earnings, which enter some specifications as controls, and to evaluate changes in adult earnings and employment among applicant households.

Criminal justice data Finally, I use court records from the Connecticut Judicial Branch to measure arrests that result in misdemeanor or felony convictions. I identify convictions from recorded guilty findings and measure the timing of these outcomes using the corresponding arrest date. I link court records to individuals in ECC application files using name and date of birth, separately from the DataLinkCT linkage. I require exact matching on date on birth and fuzzy matching on name. Conviction records are available from January 2000 to August 2024. Appendix C.3 describes these records and the matching process in more detail.

¹⁰The matching is conducted by DOL staff through the DataLinkCT initiative.

3.1 Sample construction

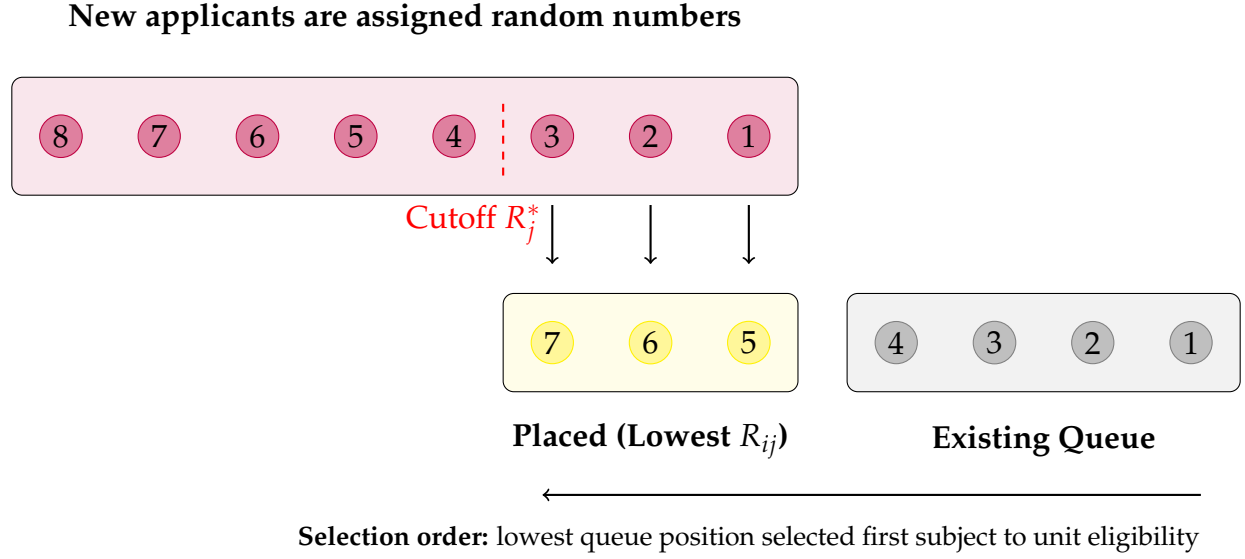
To construct the analysis sample, I begin with the universe of successful applications to ECC’s various housing assistance wait lists as of March 13, 2025, which is the sample provided by ECC for matching through DataLinkCT.¹¹ This dataset contains almost 329,000 applications to 100 different wait lists by more than 80,000 households. I make a number of restrictions to arrive at my final analysis sample.

First, because ECC indefinitely opened many of its housing assistance wait lists beginning in June 2019, I limit attention to applications from 2003-2018, prior to the indefinite opening policy and after ECC begins recording new program admissions in 2002. This restriction reduces the sample to around 75,000 applications to 81 different wait lists by more than 25,000 households. Second, I remove applications from outside of New Haven county. In Appendix Figure A.3, I report the total number of unique applicants and the share of applicants from outside New Haven County in each year from 1998-2024. Only 10.3% of applicants from 2003-2018 are removed under this restriction. I then remove specialty wait lists (e.g., the LIPH “Community Re-Entry” wait list for applicants exiting the correctional system), wait lists for developments that are restricted to elderly or disabled applicants, and wait lists for internal transfers and portability vouchers.¹² I refer to this limited set of applications as the *descriptive sample*. This sample contains more than 44,000 applications to 24 wait lists by more than 16,000 households. Around 46% of these applications are submitted by households containing children.

¹¹Successful applications are applications that result in placement onto a wait list. During wait list position lotteries, many applicants are not placed and their applications are discarded. In non-lottery periods, all applications are successful.

¹²Portability is a feature of the Housing Choice Voucher program that allows recipients to move - while retaining their voucher - to the jurisdiction of a new PHA after 12 months in the program.

Figure 2: Wait list position lotteries



Note: Figure 2 depicts the wait list assignment process. Numbers in purple are generated using a random number generator. Queue position depends on both lottery number and the size of the existing queue. Only applicants with a lottery number that falls below the predetermined lottery cutoff R_j^* are placed (1, 2, and 3 in this illustrative example). My dataset contains the *placed* (yellow) and *existing* (grey) wait list participants **but not** applicants to the left of the lottery number cutoff (4-8 in this illustrative example) for each wait list lottery j . I observe the entire household of all placed and existing applicants for each wait list opening.

Openings and lotteries I next restrict the sample to applications that occur during clearly-defined wait list opening periods. To identify wait list openings, I cross-reference monthly application counts with local news articles and ECC annual reports. I describe the process of identifying wait list openings and lotteries, which are opening periods during which applicants are placed onto the wait list in random order, in Appendix B. Figure 2 presents a stylized example of a wait list lottery, in which applicants are assigned random numbers and a subset are placed – in random order – onto an existing wait list. Using a subsample of applicants with wait list position information, I am able to identify lotteries by testing for a relationship between an applicant’s assigned position and their implied position given the exact date and time they apply to a wait list. I identify an opening as a lottery if I fail to reject the null that the correlation between these two positions, across applicants within the opening, is zero. Point estimates and 95% confidence intervals from this regression test are presented in Appendix Figure B.3.

Because my focus is on the effects of housing vouchers and public housing relative to a counterfactual of no housing assistance, I additionally exclude households living in public housing or using a voucher at baseline. The final lottery sample contains 7,399 children

from 4,194 households. Wait list lotteries occur between March 2003 and January 2018 across 9 opening periods and 14 wait lists.¹³ I summarize the number and characteristics of children in each lottery in Appendix Table A.1.

Matching I work with DataLinkCT and ECC to match wait list applications to employment records from DOL and education records from SDE using name, date of birth, and the last 4 digits of the Social Security number. I refer to the *analysis sample* as the set of lottery sample applications for which I match a child to any SDE record. The matched SDE analysis sample contains 5,590 children from 3,480 households (the child-level match rate is 75.6%). The matched SDE and DOL linked sample contains 5,505 children from 3,417 households. My sample includes all outcomes for students that enroll in any New Haven county school in any school year, so I observe educational outcomes for students who exit New Haven county but remain in Connecticut.¹⁴ I fail to observe educational outcomes for children who never attend a public school in New Haven county (e.g., the child of an applicant who moves to Bridgeport after the lottery). My main results in Section 5 use different subsets of the matched SDE sample, depending on child age at the time the outcome is measured.¹⁵ Finally, I confirm that children in households that are selected from a wait list are not more likely to match to the administrative records. In Appendix Table A.2, I show that match rates are nearly identical ($p = 0.305$) for children in offered and not offered households after adjusting for differences in child characteristics. Unadjusted, match rates are actually lower for offered children. Table 1 reports summary statistics across the descriptive, lottery, and matched samples. Applicant children are similar along observable dimensions across samples.

¹³Two opening periods contain multiple site-specific wait lists.

¹⁴I include a map of Connecticut counties in Appendix Figure A.5.

¹⁵My Data Use Agreement (DUA) with DataLinkCT does not allow me to observe child age in the matched sample. I impute age from information on child grade derived from test score and school attendance records under the assumption that each child is 5 at the time they first enroll in kindergarten.

Table 1: Applicant child summary statistics

	Descriptive	Lottery		
		All	SDE	SDE & DOL
Panel A: Child and household characteristics				
Black	0.55	0.57	0.58	0.58
Female	0.49	0.49	0.49	0.49
Age	5.62	5.68	5.12 [†]	5.11 [†]
Household size	3.89	3.79	3.79	3.81
Household head is citizen	0.50	0.51	0.54	0.54
Household head is disabled	0.09	0.06	0.07	0.07
Household head age	31.09	30.44	–	–
Household earnings (2023 dollars)	–	–	–	\$16,524.87
Household employment	–	–	–	0.74
Panel B: Assignment outcomes				
Offered by age 18	0.62	0.61	0.61	0.61
Assisted by age 18	0.30	0.26	0.30	0.30
Wait time (yrs)	3.63	4.13	4.16	4.17
Panel C: Sample counts				
# Children	13,815	7,399	5,590	5,505
# Applications	8,164	4,328	3,587	3,524
# Wait Lists	24	14	14	14
# Households	7,565	4,194	3,480	3,417

Notes: Panel A of Table 1 presents averages for the children of applicants to ECC wait lists. Age is measured for each child at the time of application. Earnings is total household earnings reported by DOL in the year prior to the application year (measured in 2023 dollars). Employed is an indicator equal to 1 if household earnings in the year prior to application are non-zero. Panel B of Table 1 reports the share of children in each sample in households that receive interview offers or housing assistance by age 18 (or the last recorded age). Average wait time is the average time between the child's earliest application and their earliest date of program participation, reported conditional on receiving assistance. The data source is ECC wait list application records from 2003-2018 and DOL earnings records from 2001-2021. The SDE column includes all households with at least one child aged 0-17 at the time of application who matches to the SDE records. I impute age from the earliest observed child grade in these columns, which are marked with a †. The SDE & DOL column is limited to households with at least one adult who matches to the DOL records. DOL and SDE matches require exact matching.

4 Empirical strategy

4.1 Instrumental variables specification

I use variation in when households reach the top of lottery-ordered waiting lists to estimate the effects of housing assistance on children. Let i index children, j index wait list openings, and g denote the child's age when an outcome is measured. Let Y_{ijg} denote an outcome for child i , whose household applied during opening j , measured at age g .

Define $Z_{ijg} = 1$ if the household has been selected from the wait list for an eligibility interview before the outcome is measured at age g . I refer to this selection as an offer. Offer status is absorbing: once a household has been selected from list j , Z_{ijg} remains equal to one at all future ages. Define $D_{ijg} = 1$ if the household has received housing assistance at any point between opening j and the measurement of the outcome at age g . Like the offer instrument, treatment is absorbing, and a household that enters and subsequently exits housing assistance remains treated.

For each age group, I estimate the following first and second-stage equations by two-stage least squares:

$$D_{ijg} = \pi_b + \delta_b Z_{ijg} + X'_{ijg} \Psi_b + \lambda_{j,s(i)} + v_{ijg}, \quad (1)$$

$$Y_{ijg} = \alpha_b + \beta_b D_{ijg} + X'_{ijg} \Gamma_b + \lambda_{j,s(i)} + \varepsilon_{ijg}, \quad (2)$$

where $b = b(g)$ denotes the age bin containing age g . The fixed effects $\lambda_{j,s(i)}$ are defined by interacting wait list opening j with the household-size-by-disability-status cell $s(i)$. These interacted fixed effects account for differences across openings and the administrative rules that match households to vouchers or available units based on household size and disability status. The vector X_{ijg} contains predetermined child and household characteristics, including race, gender, and age at application, as well as the outcome year. Standard errors are clustered at the household level to allow for dependence across siblings and repeated observations of the same child at different ages.

I also estimate the reduced-form effect of an offer by replacing D_{ijg} in Equation (2) with Z_{ijg} . The reduced-form estimate measures the effect of being selected from the wait list by age g , while δ_b measures the first-stage effect of selection on assistance. In an exactly identified specification, the 2SLS coefficient β_b scales the reduced-form effect by the corresponding first stage. I report the reduced form, first stage, and 2SLS estimate using the same sample and controls.

For annual outcomes constructed from school records, I sometimes stack child-year ob-

servations within the two-year age bins 6–7, 8–9, 10–11, 12–13, 14–15, and 16–17 and estimate Equations (1)–(2) separately in each bin. For other outcomes, the age bin equals the child age. The coefficient β_b allows the effect of entering assistance to vary with the age at which the outcome is measured. For outcomes measured at a terminal age, such as high school graduation by age 20, I use one observation per child and estimate a separate specification for each outcome and terminal age. In the attainment specifications, $D_{ij,18}$ indicates any assistance receipt through age 18 and $Z_{ij,18}$ indicates any offer through age 18, even when the outcome is measured after age 18.

4.2 Identification and interpretation

A causal interpretation of β_b requires four conditions. First, conditional on the $\lambda_{j,s(i)}$ fixed effects and included controls, realized offer timing must be independent of households’ potential assistance histories and children’s potential outcomes. While initial position is randomly assigned within the lottery openings, the instrument also depends on voucher or unit turnover and household eligibility for particular units. A conditional independence assumption additionally requires that administrative processing, housing availability, and ECC list processing do not select households based on unobserved determinants of children’s outcomes after conditioning on unit eligibility.¹⁶

Second, wait list selection must have a nonzero effect on assistance receipt. I assess this relevance condition by evaluating the first stage directly for each estimation sample. Third, I require monotonicity: selection from the wait list by the relevant cutoff cannot cause households to select out of assistance. Fourth, the exclusion restriction requires that selection affect children’s outcomes only through their housing assistance history. This rules out direct effects of the orientation interview, housing search, anticipation of assistance, or access to other services associated with the offer. It also requires that one household’s position or receipt not directly affect the outcomes of children in other applicant households.

Under these assumptions, β_b identifies the local average treatment effect of receiving housing assistance by age $b(g)$ for households induced to receive assistance by wait list

¹⁶A useful benchmark is $Z_{ijg} = \mathbf{1}\{R_{f(i)j} \leq K_{j,s(f(i)),t(g)}\}$, where $f(i)$ denotes child i ’s household, $R_{f(i)j}$ is the household’s randomly assigned position in opening j , $s(f(i))$ is its baseline household-size-by-disability-status cell, and $K_{j,s,t}$ is the largest position processed from that cell by date t . If $K_{j,s,t}$ is determined by voucher or unit availability and administrative capacity before applicant responses are observed, then, conditional on j , s , and t , offers depend only on random position. In practice, ECC may extend the cutoff when earlier applicants do not respond, fail eligibility verification, or decline assistance. Realized offers are then not exactly randomized. I treat the resulting dependence as negligible under a large wait list approximation (De Chaisemartin and Behaghel 2020).

selection. Because D_{ijg} indicates any assistance by age g , I capture the effect of gaining access to assistance, including the timing and duration of the resulting assistance spell.

4.3 Cumulative exposure specifications

For some long-run outcomes, I replace the binary treatment in Equations (1)–(2) with the number of years a child has received assistance before the relevant outcome and instrument it with the number of years since the household’s first offer. These estimates allow me to exploit the total variation in exposure to assistance generated by offer timing. Interpreting the resulting coefficient as the causal effect of one additional assisted year requires stronger restrictions than the binary entry specification: offer timing must weakly increase the number of years with assistance, and the effect of each year with assistance must be constant. This “dosage” model assumes that the impacts of assistance do not depend on the child’s age at receipt. Some studies find support for this assumption in similar contexts (Chyn 2018; Chaudhry and Eng 2024; Beauregard 2025; Ribeiro and Mariante 2025). I treat the cumulative exposure estimates as a complement to, rather than a substitute for, the binary treatment design.

4.4 Balance and first stage

I begin by validating the research design, checking that the realized timing of offers is unrelated to households’ economic circumstances or labor market trajectories before application. In practice, I test whether household offer status at different ages is related to pre-application household employment and earnings or child demographics.

For each age bin b , I use the same sample and offer instrument Z_{ijg} as in Equations (1)–(2). For each predetermined characteristic X_i^k , I estimate

$$X_i^k = \alpha_b^k + \beta_b^k Z_{ijg} + A'_{ijg} \rho_b^k + \lambda_{j,s(i)}^k + \varepsilon_{ijg}^k, \quad g \in b, \quad (3)$$

where A_{ijg} contains application age, calendar year, and years since lottery fixed effects, and $\lambda_{j,s(i)}^k$ are the lottery-by-size-by-disability fixed effects, which control for characteristics used by ECC to match applicants to available units. I cluster standard errors at the household level.

I check for balance based on child gender and race, household employment and earnings in the year before application, and changes in household employment and earnings over

the two years preceding application. I report the coefficient on offer status for each characteristic and the p -value from a joint test that all balance coefficients are zero in Table 2. While pre-application changes in earnings are marginally significant at older ages, I fail to reject the joint null that all coefficients are equal to zero at any age. The results are similar if I examine balance in the timing of waiting list selection after a lottery.

Table 2: Offer timing balance by child age

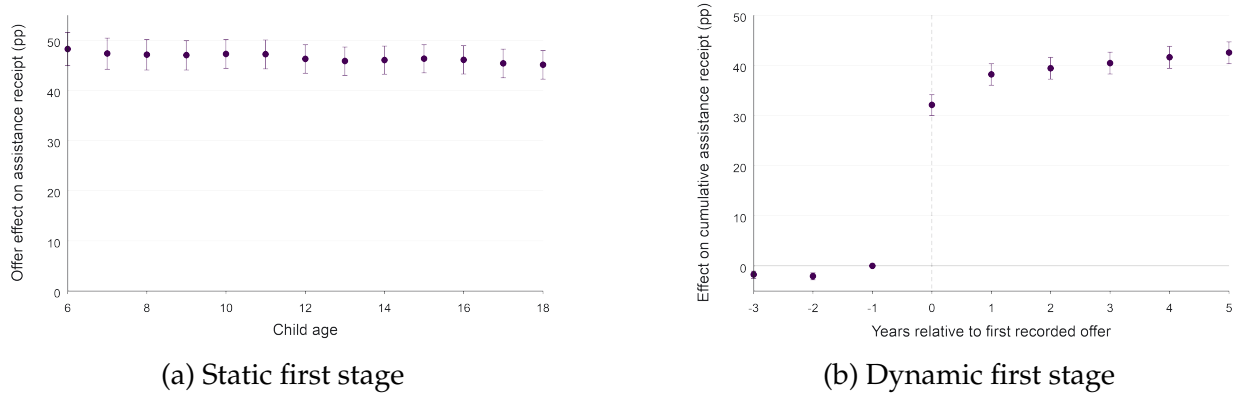
	$Z_{ijg} = 1\{\text{Selected from list by age } g\}$					
	Ages 6 – 7	Ages 8 – 9	Ages 10 – 11	Ages 12 – 13	Ages 14 – 15	Ages 16 – 17
Black	0.007 (0.020)	0.003 (0.019)	-0.004 (0.019)	-0.005 (0.020)	-0.004 (0.020)	0.003 (0.020)
Female	0.000 (0.017)	-0.010 (0.016)	-0.005 (0.015)	-0.009 (0.014)	-0.011 (0.014)	-0.006 (0.014)
Any adult employed, prior year	-0.017 (0.019)	-0.011 (0.019)	-0.010 (0.019)	-0.010 (0.019)	-0.010 (0.019)	-0.006 (0.019)
Change in employment, years –2 to –1	-0.009 (0.019)	-0.012 (0.019)	-0.014 (0.019)	-0.019 (0.019)	-0.014 (0.019)	-0.016 (0.019)
Household earnings, prior year (\$1,000)	0.517 (0.738)	0.260 (0.735)	0.126 (0.754)	-0.032 (0.811)	-0.155 (0.833)	-0.197 (0.844)
Change in earnings, years –2 to –1 (\$1,000)	0.575 (0.454)	0.673 (0.459)	0.850* (0.472)	0.885* (0.485)	0.854* (0.499)	0.927* (0.507)
Joint test p -value	0.467	0.462	0.260	0.214	0.344	0.369
Child-year observations	7,622	8,856	9,953	10,856	11,548	12,098
Children	3,995	4,573	5,102	5,525	5,850	6,111
Households	2,729	2,948	3,120	3,241	3,330	3,390
Share offered	0.469	0.513	0.546	0.570	0.588	0.598

Notes: Table 2 reports conditional balance tests for the binary offer instrument Z_{ijg} . The sample excludes households receiving assistance before the lottery. Each entry is the coefficient from a separate regression of the listed predetermined characteristic on an indicator for selection from a wait list by the age at which the observation is measured. All specifications include lottery and calendar year age, calendar year, and lottery-by-household-size-by-disability-status fixed effects. Standard errors are clustered at the household level and appear in parentheses. Earnings variables measure W-2 earnings and are reported in thousands of dollars. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Relevance Reaching the top of a lottery-ordered wait list strongly predicts subsequent housing assistance receipt, as expected. Figure 3 shows the static and dynamic relationship between offer timing and the receipt of assistance. Because the offer represents selection from the top of the wait list, incomplete take-up is driven by a combination of non-response, ineligibility, and failure to lease up. In a similar setting, Waldinger 2021 documents significant non-response to notifications on housing assistance wait lists. In the dynamic event study (Sun and Abraham 2021), the probability of receiving assistance after an offer rises slightly over time to over 40 percentage points at year five.

Table 3 reports corresponding first-stage estimates of Equation (1) for each age two-year age bin. Being offered by age bin b raises the probability of receiving assistance by roughly

Figure 3: Wait list selection first stage



Note: Figure 3 plots the first stage relationship between reaching the top of a wait list and subsequent receipt of housing assistance. Figure 3a measures assistance at each age from six to eighteen. Figure 3b measures the dynamic effect of being selected from a wait list on being assisted, using the method in Sun and Abraham 2021. Whiskers report 95% confidence intervals. The data source is the lottery sample of ECC applications from 2003-2018.

45 percentage points in the matched sample, with first stage F -statistics above 900 across ages. Being selected from a list has a sizable effect on receiving assistance.

Table 3: Wait list selection first stage by child age

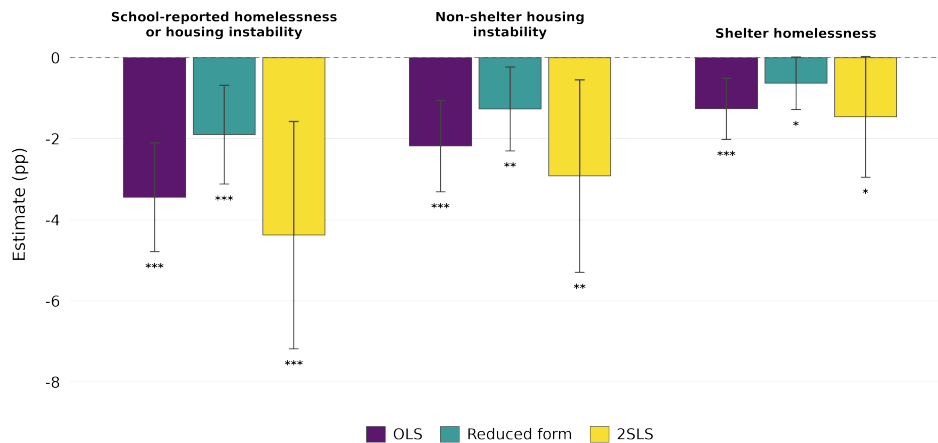
	Outcome: assistance received by age b , D_{ijb}					
	Ages 6–7	Ages 8–9	Ages 10–11	Ages 12–13	Ages 14–15	Ages 16–17
Offer indicator (Z_{ijb})	0.476*** (0.016)	0.468*** (0.015)	0.469*** (0.014)	0.456*** (0.014)	0.457*** (0.014)	0.452*** (0.014)
Outcome mean	0.24	0.26	0.28	0.29	0.29	0.30
Demographic controls	Y	Y	Y	Y	Y	Y
Size $\times$ disability fixed effects	Y	Y	Y	Y	Y	Y
Age and timing fixed effects	Y	Y	Y	Y	Y	Y
Adjusted R^2	0.40	0.39	0.39	0.38	0.37	0.37
First stage F -statistic	912.4	983.3	1,060.5	998.0	1,022.1	976.9
# Observations	6,988	8,087	9,059	9,829	10,360	10,769
# Applications	2,748	2,985	3,193	3,328	3,425	3,491
# Children	3,663	4,167	4,639	4,991	5,239	5,431
# Households	2,643	2,879	3,086	3,221	3,318	3,384

Notes: Table 3 present first stage estimates using the instrument Z_{ijb} , is an indicator for whether the household had received an offer by that age bin. The sample is the matched sample of children linked to SDE records with any DOL household link. Specifications include race and sex controls and fixed effects for age at application, age, outcome year, years since lottery, and lottery $\times$ family size $\times$ disability-status cells. Standard errors are clustered at the household level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

5 Short-run outcomes

5.1 Effects on housing stability

Figure 4: Effects on school-reported housing instability



Note: Figure 4 plots pooled ages six to eleven estimates for school-reported homelessness or housing instability, non-shelter housing instability, and shelter homelessness. The outcomes are constructed using McKinney-Vento records. The figure reports OLS estimates (purple), reduced-form estimates of being offered assistance by a given post-lottery age (green), and 2SLS estimates of receiving assistance by that age (yellow), instrumented by an offer by the same year. Estimates pool child observations from ages six to eleven. The y-axis is in percentage points. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the household level. Whiskers represent 95% confidence intervals.

Having assessed the validity of the offer instrument, I begin by using the research design in Equations (1)-(2) to explore effects on the child outcomes most directly related to housing assistance. Pooling annual school observations from ages six through eleven, receiving housing assistance by a given age reduces school-reported homelessness or housing instability by 4.38 percentage points ($p = 0.002$), relative to a complier control mean of 6.56 percent. Pooled estimates are reported in Figure 4. The binned estimates in Table 4 show that the reduction is largest among younger children. At ages six to seven, assistance reduces school-reported homelessness by 7.29 percentage points ($p < 0.001$), relative to a complier control mean of 9.16 percent. The estimate is 3.38 percentage points at ages eight to nine ($p = 0.082$) and becomes smaller and less precise at older ages.

The estimated reductions are concentrated in the forms of school-reported instability most common in the sample. Pooling observations from ages six through eleven, assistance reduces non-shelter housing instability by 2.92 percentage points ($p = 0.016$) and

Table 4: Effects of housing assistance on school-reported homelessness by child age

Child ages	OLS	RF	2SLS	CM	CCM	Observations	# Children	First stage F
6–7	-4.14*** (0.87)	-3.30*** (0.81)	-7.29*** (1.79)	5.20 –	9.16 (1.73)	5,317 –	2,932 –	765.0 –
8–9	-3.68*** (0.94)	-1.44* (0.83)	-3.38* (1.94)	4.10 –	5.58 (1.83)	6,119 –	3,318 –	779.7 –
10–11	-2.49*** (0.88)	-0.94 (0.76)	-2.24 (1.82)	3.39 –	4.79 (1.68)	6,607 –	3,568 –	766.7 –
12–13	-2.22*** (0.78)	-1.14 (0.70)	-2.92 (1.80)	3.59 –	4.54 (1.69)	6,610 –	3,580 –	584.8 –
14–15	-1.38* (0.80)	-0.75 (0.67)	-2.00 (1.82)	3.44 –	4.84 (1.65)	6,295 –	3,413 –	517.1 –
16–17	-1.38 (0.97)	-0.58 (0.97)	-1.61 (2.72)	3.55 –	4.59 (2.48)	4,120 –	2,772 –	331.4 –

Notes: Each row reports an estimate of β_b after stacking annual child-year observations within the indicated two-year age bin. Coefficient estimates and means are in percentage points. OLS reports the coefficient on receiving assistance by the corresponding age. RF denotes the reduced-form effect of having been offered assistance by that year. The binary 2SLS specification instruments contemporaneous assistance receipt with the offer indicator. CM is the mean outcome among observations not yet offered assistance. CCM is the estimated outcome mean among untreated compliers. Standard errors are shown in parentheses below coefficient estimates and CCMs. Specifications include fixed effects for age-at-application, current age, calendar year, years since application, and lottery-by-family-size-by-disability allocation cells, in addition to controls for child race and gender. Standard errors are clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, respectively.

doubled-up homelessness by 2.91 percentage points ($p = 0.008$). The estimated effect on shelter homelessness is also negative, at 1.46 percentage points ($p = 0.055$). I find no detectable effect on hotel or motel homelessness (0.10 percentage points, $p = 0.829$) or unsheltered homelessness (-0.11 percentage points, $p = 0.315$), both of which are rare in the SDE records. In other words, the aggregate decline in school-reported housing instability largely reflects reductions in less visible forms of housing instability – particularly doubled-up homelessness – that are recorded under the McKinney–Vento Act but generally do not appear in other administrative datasets used to study homelessness. This pattern is consistent with studies finding large effects of housing interventions on these wider measures of housing instability (Collinson and Gibbs 2025; Liao et al. 2025).

5.2 Effects on early childhood development

Since 2010, Connecticut has administered the Kindergarten Entrance Inventory (KEI) to children entering public kindergarten. Teachers assess children across six categories: language, literacy, numeracy, creativity, physical and motor development, and personal and social development. Scores range from 1 (needs substantial instructional support) to 3 (consistently demonstrates the skill). I standardize each category within each school year. For some outcomes, I group language, literacy, and numeracy together as academic skills and creativity, physical and motor development, and personal and social development together as developmental skills.

Averaged across all children, the estimated effect on the overall KEI index is small and imprecise, at 0.064 standard deviations ($p = 0.499$). This pooled estimate, however, hides a more favorable pattern for boys, for whom receiving housing assistance before kindergarten increases the average of the development skill measures – creativity, physical and motor development, and personal and social development – by 0.31 standard deviations ($p = 0.056$). A latent factor measurement model that separates these scores into noisy measures of development and academic skill produces a similar estimate: assistance increases boys’ latent developmental skill by 0.35 standard deviations ($p = 0.040$). In contrast, the estimated effect on boys’ latent academic skill is small and statistically insignificant. Additional details on the latent factor model are available in Appendix D. The corresponding latent developmental skill estimate for girls is 0.01 standard deviations ($p = 0.930$), though the estimated difference between boys and girls is imprecise ($p = 0.156$). I report the estimates for the standardized category outcomes in Table 5 and the measurement model estimates for the developmental factor in Appendix Table A.3.

Table 5: Effects on kindergarten readiness, by gender

Outcome	Pooled	Boys	Girls
KEI index	0.064 (0.094)	0.206 (0.130)	0.023 (0.136)
<i>Developmental domains</i>			
Creativity/aesthetic	0.117 (0.111)	0.306* (0.160)	0.003 (0.157)
Physical/motor development	0.045 (0.110)	0.231 (0.159)	−0.043 (0.156)
Personal/social development	0.166 (0.111)	0.328** (0.158)	0.135 (0.160)
<i>Academic domains</i>			
Numeracy	0.054 (0.112)	0.083 (0.156)	0.094 (0.163)
Literacy	0.040 (0.108)	0.173 (0.151)	−0.003 (0.155)
Language	−0.037 (0.110)	0.115 (0.155)	−0.044 (0.159)
Children	2,118	1,078	1,039
Households	1,735	972	953

Notes: Table 5 reports binary 2SLS estimates for the effects of housing assistance on early childhood development. The endogenous treatment is receipt of housing assistance by the kindergarten assessment year, instrumented with an offer by that year. The KEI index averages standardized language, literacy, numeracy, creativity/aesthetic, physical/motor, and personal/social scores. Each score is standardized within a given school year. KEI scores are available beginning in the 2008-2009 school year. Standard errors are clustered at the household level. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$, respectively.

5.3 Effects on elementary and middle school test scores

Despite sizable reductions in school-reported housing instability and improvements in early childhood development among boys, I find no consistent evidence that housing assistance improves measured academic achievement in elementary and middle school. In the binary specification, pooled effects on end-of-grade test scores in grades 3–6 are all positive, ranging from 0.028 to 0.159 standard deviations, but none is statistically significant. Estimates in grades 7–8 are smaller or negative, ranging from -0.214 to 0.023 standard deviations, and are also imprecise. I typically cannot rule out large effect sizes, especially at younger ages. I do find a large, significant 0.446 standard deviation increase in English language arts scores for boys in third grade ($p = 0.038$), with a similar, marginally significant estimate in fifth grade.

The absence of positive achievement effects in elementary and middle school places these New Haven estimates closer to the findings from the Chicago housing voucher lottery studied by [Jacob et al. 2015](#) than to the positive elementary school effects documented in the Miami-Dade housing lottery studied by [Collinson and Gibbs 2025](#). At the same time, the positive developmental estimates for boys suggest that standardized achievement tests may not capture all of the skills affected by increased housing stability. One possible explanation for the absence of significant achievement gains is that the New Haven lotteries improved housing stability without systematically moving children into higher-performing schools. Consistent with this interpretation, the school quality estimates in Section 5.4 below show no improvements in school value added and some evidence that assistance moved older children into schools serving more economically disadvantaged students.

5.4 Other outcomes

The preceding results show that housing assistance reduces housing instability and improves early developmental skills among boys without producing significant improvements in test scores in elementary and middle school. This pattern suggests that assistance may affect children’s home environments and their attachment to school without substantially changing other inputs. I next examine three potential channels and intermediate outcomes: school attachment, household resources, and school quality.

School attachment I find evidence that housing assistance improves children’s attachment to school, particularly during elementary school. Pooling child-year observations from ages 6–11, receiving assistance increases attendance by 1.14 percentage points ($p =$

0.027) and reduces chronic absenteeism by 6.46 percentage points ($p = 0.028$). Assistance also reduces within-year school switching by 2.70 percentage points ($p = 0.014$). The effect on switching school districts is also negative but smaller and statistically insignificant, at -1.37 percentage points ($p = 0.153$). Together, these results are consistent with a school continuity channel for housing assistance at early ages, keeping children more consistently attached to school despite no evidence of significant improvements in academic achievement. Recent studies find that increased school attendance has a positive effect on longer-run outcomes (Liu et al. 2021; Goldman and Gracie 2024).

Table 6: Effects of housing assistance on school continuity at ages 6–11

Outcome	OLS	RF	2SLS	CM	CCM	Observations	# Children	First stage F
Attendance rate	-0.13 (0.26)	0.49** (0.22)	1.14** (0.52)	92.36 –	90.67 (0.46)	18,043 –	4,110 –	946.4 –
Chronic absenteeism	-0.31 (1.45)	-2.80** (1.26)	-6.46** (2.93)	25.24 –	34.06 (2.69)	18,043 –	4,110 –	946.4 –
Switched schools within year	-1.76*** (0.52)	-1.17** (0.48)	-2.70** (1.10)	5.05 –	6.04 (1.04)	18,043 –	4,110 –	946.4 –
Switched districts within year	-1.84*** (0.44)	-0.60 (0.42)	-1.37 (0.96)	3.40 –	3.63 (0.91)	18,043 –	4,110 –	946.4 –

Notes: Table 6 reports OLS, reduced form, and 2SLS estimates of the effects of housing assistance on school continuity. The sample pools child-year observations from ages 6 through 11. Coefficient estimates and means are in percentage points. OLS reports the coefficient on receiving assistance by the corresponding child age. RF denotes the reduced form effect of having received an offer by that year. The 2SLS specification instruments assistance receipt with the offer indicator. CM denotes the mean among observations not yet offered assistance, and CCM denotes the estimated complier control mean. Standard errors are shown in parentheses below coefficient estimates and CCMs and are clustered at the household level. Specifications include controls for child characteristics and fixed effects for age at application, age, calendar year, years since application, and lottery-by-family-size-by-disability-status. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Household resources Housing assistance may affect children by changing both the time and financial resources available in the household. In particular, rent subsidies can alter adults’ incentives to work, and existing studies of housing assistance lotteries find evidence of average reductions in adult employment or earnings (Jacob and Ludwig 2012; van Dijk 2019). I examine this channel using information on household employment and earnings from W-2 records in Connecticut.

I find no evidence that housing assistance reduces adult employment or earnings. Instead, the estimated employment effects are positive in every child age bin but are generally imprecise. Estimated effects on household earnings are also positive and tend to increase with children’s ages, though the estimates through age 15 are statistically insignificant. I report the full set of estimates in Appendix Table A.5. Altogether, the estimates provide relatively clear evidence against an adverse adult labor supply response

but only suggestive evidence that increased earnings are an important channel for any early childhood effects.

School quality Housing assistance may also affect children by changing the quality of the schools they attend. For elementary and middle schools, I measure school quality using school-by-year value added in ELA and math, estimated from a lagged achievement model, as well as school proficiency rates and student characteristics. Appendix C.1.4 describes the value added model and presents the distribution of the school-by-year value added estimates.¹⁷

I find no evidence of effects on school value added. But older students that receive assistance do appear to attend schools with different peers. Receiving assistance increases the share of students eligible for free or reduced-price lunch at the child's school by 5.92 percentage points at ages 14–15 ($p = 0.008$) and by 7.21 percentage points at ages 16–17 ($p = 0.012$). I find no effects at younger ages. This pattern is closer to the limited changes in school quality found in the Chicago voucher lottery (Jacob et al. 2015) than to settings in which housing assistance induces moves to substantially higher-opportunity environments. I report the full set of school quality estimates in Appendix Table A.6.

6 Long-run outcomes

The preceding section describes how housing assistance affects children's housing stability, skill development, and school environments. Now, I examine whether these changes translate into differences in educational attainment and criminal justice involvement as children enter adulthood.

6.1 Effects on educational attainment

In contrast to the test score results discussed above, housing assistance has large positive effects on educational attainment. I report the full set of estimates in Table 7 using two complementary specifications. The binary specification instruments receiving any assistance by age 18 with an indicator for receiving an offer by age 18. The exposure specification instead instruments years of assistance through age 18 with years since an offer. Under the constant marginal effects assumption discussed in Section 4, the latter coefficient can be interpreted as the effect of one additional year of childhood assistance

¹⁷Because the administrative records do not identify teachers, the value added measures combine school effects with the effects of teacher sorting across schools or student sorting across classrooms.

(Chyn 2018; Chaudhry and Eng 2024). Because this interpretation is stronger than the assumptions required for the binary design, I report both estimates and treat the exposure results as a complement to the binary estimates.

The high school graduation estimates are positive but imprecisely estimated. An additional year of assistance increases on-time graduation by 1.42 percentage points ($p = 0.180$) and graduation by age 20 by 1.93 percentage points ($p = 0.151$). Among children old enough to be observed through age 22, the estimate rises to 2.88 percentage points ($p = 0.099$), relative to a control mean of 63.1 percent. The corresponding binary estimates are also positive, but none is statistically significant at the 5 percent level. These results are suggestive of increases in high-school completion, especially at later ages, which is consistent with estimates of the effects of eviction on child educational attainment (Collinson et al. 2025).

The evidence is stronger for postsecondary enrollment. An additional year of assistance increases any college enrollment by age 19 by 1.94 percentage points ($p = 0.079$), by age 20 by 3.02 percentage points ($p = 0.022$), and by age 22 by 4.19 percentage points ($p = 0.020$). The age twenty two effect is concentrated in four-year enrollment: an additional year of assistance raises four-year enrollment by 3.92 percentage points ($p = 0.017$), compared with a smaller and statistically insignificant 1.29 percentage point effect on two-year enrollment ($p = 0.412$). I also find suggestive effects on degree completion. The estimated effect on college graduation is 1.99 percentage points by age 22 ($p = 0.054$) and 2.92 percentage points by age 24 ($p = 0.085$).

This pattern – larger effects on attainment than on standardized tests – is similar to findings in other papers that study the effects of in-kind transfers in early childhood on long-run outcomes (Deming 2009; Gray-Lobe et al. 2023). For example, Gray-Lobe et al. 2023 find that Boston public preschool increased high school graduation by 6.0 percentage points and college enrollment by 5.4 percentage points despite having no detectable effect on state achievement tests. Within the housing literature, the positive estimates differ from the largely null child outcome results in the Chicago voucher lottery studied by Jacob et al. 2015, although the designs and treatment margins are not directly comparable: that study estimates the effect of ever using a voucher, whereas my preferred dose-response specification scales the effect by years of assistance. More broadly, the results are consistent with evidence that longer childhood exposure to improved housing environments can affect adult outcomes (Chyn 2018; Chaudhry and Eng 2024).

Table 7: Housing assistance effects on high school and college outcomes

Outcome	CM	Children	Households	Exposure		Binary	
				2SLS	First stage F	2SLS	First stage F
Panel A: High school graduation							
HS graduation by age 18	53.6	3,057	2,096	1.42	222.8	16.34*	268.2
	–	–	–	(1.06)	–	(9.00)	–
HS graduation by age 20	62.4	2,358	1,656	1.93	154.8	14.91	196.8
	–	–	–	(1.34)	–	(10.22)	–
HS graduation by age 22	63.1	1,727	1,252	2.88*	114.5	10.74	146.2
	–	–	–	(1.75)	–	(12.11)	–
Panel B: College enrollment and completion							
College enrollment by age 19	34.0	2,685	1,853	1.94*	196.7	15.38*	227.5
	–	–	–	(1.10)	–	(9.18)	–
College enrollment by age 20	35.1	2,358	1,656	3.02**	154.8	19.07*	196.8
	–	–	–	(1.32)	–	(10.23)	–
College enrollment by age 22	38.2	1,727	1,252	4.19**	114.5	14.53	146.2
	–	–	–	(1.80)	–	(12.42)	–
Four-year college enrollment by age 22	22.1	1,727	1,252	3.92**	114.5	14.73	146.2
	–	–	–	(1.65)	–	(10.87)	–
Two-year college enrollment by age 22	23.2	1,727	1,252	1.29	114.5	3.40	146.2
	–	–	–	(1.58)	–	(10.65)	–
College degree by age 22	6.6	1,727	1,252	1.99*	114.5	10.42	146.2
	–	–	–	(1.03)	–	(6.56)	–
College degree by age 24	9.1	1,173	897	2.92*	75.9	19.44**	102.4
	–	–	–	(1.70)	–	(9.86)	–

Notes: Table 7 reports 2SLS estimates of the effect of housing assistance on high school and college outcomes using offer timing as an instrument. All coefficient estimates and means are in percentage points. Exposure columns report the effect of one additional year of assistance through age 18, instrumenting assisted years with offered years. Binary columns report the effect of any assistance through age 18, instrumenting assistance receipt with an offer. CM is the outcome mean among children whose households were not offered assistance by age 18. Standard errors, shown in parentheses, are clustered at the household level. Specifications include controls for child and household characteristics and fixed effects for age at application, outcome year, and lottery-by-family-size-by-disability-status cells. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

6.1.1 Heterogeneity in attainment effects

Table 8 reports the dose-response estimates separately by child gender. The point estimates for any college enrollment and college completion are especially large for females relative to males, providing suggestive evidence that gains in postsecondary attainment may be concentrated among females.

Table 8: Attainment effects by child gender

Outcome	Male	Female	Female – male	<i>p</i> -value
High school graduation by age 22	3.47 (2.32)	2.31 (2.94)	-1.17 (3.78)	0.758 –
Any college enrollment by age 22	1.42 (2.21)	8.65** (3.57)	7.23* (4.26)	0.090 –
Four-year college enrollment by age 22	2.76 (1.78)	5.73* (3.20)	2.97 (3.63)	0.413 –
College degree by age 24	0.61 (1.49)	6.13 (3.84)	5.51 (4.15)	0.184 –

Notes: Table 8 reports 2SLS estimates, in percentage points, of the effect of one additional year of housing assistance through age 18. Assisted years are instrumented with offered years. The final two columns report the female-minus-male difference and its two-sided *p*-value from a pooled interaction specification. Standard errors, shown in parentheses, are clustered at the household level. The age twenty two samples contain 870 male and 857 female children. The age twenty four samples contain 589 male and 584 female children. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

In Table 9, I decompose exposure into years of voucher assistance and years of unit-based assistance and instrument them jointly with the corresponding offer-exposure measures. The voucher estimates are positive for all four outcomes, while the unit-based estimates are close to zero or negative. This provides suggestive evidence that educational attainment gains are concentrated among voucher recipients.

6.2 Effects on crime

I next examine whether childhood housing assistance affects criminal justice involvement in early adulthood. I separately link children in the wait list sample to Connecticut criminal court records and construct an indicator for any post-lottery docket with an explicit guilty finding on a felony charge by ages 19 through 22. Outcomes are dated using the age at arrest. Because the court extract excludes sealed, erased, pardoned, and otherwise non-disclosable cases, this outcome measures only recorded felony convictions.

I estimate these effects using the full sample of applicants from New Haven County, re-

Table 9: Attainment effects by housing assistance program

Outcome	Voucher	Unit-based	Unit – voucher	<i>p</i> -value
High school graduation by age 22	3.65** (1.50)	-1.54 (1.77)	-5.19** (2.10)	0.014 –
Any college enrollment by age 22	4.54*** (1.56)	0.00 (1.85)	-4.54** (2.16)	0.036 –
Four-year college enrollment by age 22	3.50** (1.40)	0.66 (1.67)	-2.84 (1.95)	0.146 –
College degree by age 24	2.39* (1.34)	0.03 (1.52)	-2.36 (1.72)	0.169 –

Notes: Table 9 reports program-specific 2SLS estimates in percentage points. The endogenous variables are years of voucher assistance and years of unit-based assistance through age 18, jointly instrumented by years since a voucher offer and years since a unit offer. The final two columns report the unit-minus-voucher difference and its two-sided *p*-value. Standard errors, shown in parentheses, are clustered at the household level. The age twenty two sample contains 1,727 children in 1,252 households. The age twenty four sample contains 1,173 children in 897 households. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

ardless of SDE or DOL match.¹⁸ Receiving assistance by age 18 reduces the probability of a felony conviction by age 20 by 3.88 percentage points ($p = 0.190$) and by age 22 by 7.45 percentage points ($p = 0.097$). The final estimate is large relative to the estimated 8.86 percent conviction rate among untreated compliers. The exposure estimates follow a similar but imprecise pattern, with the age twenty two estimate suggesting an annual reduction in felony conviction risk of 1.07 percentage points ($p = 0.175$).

The conclusion is somewhat stronger, given the larger sample size, when I expand the geographic sample from New Haven County applicants to all applicants from Connecticut. In this larger sample, receiving assistance by age 18 reduces felony convictions by age 22 by 9.76 percentage points ($p = 0.045$), while an additional assisted year reduces convictions by 1.34 percentage points ($p = 0.097$). Earlier estimates are also negative but less precise. Overall, the results provide evidence that childhood housing assistance reduces serious criminal justice involvement by early adulthood. They complement prior evidence that housing and neighborhood interventions can affect later criminal involvement (Kling et al. 2007; Chyn 2018).

6.3 Robustness

I now check whether the main results are sensitive to the treatment definition. I first show that the main conclusions are qualitatively similar when assistance and offer exposure are

¹⁸I am unable to combine this sample with the linked administrative records from DataLinkCT because of data privacy concerns.

instead calculated through age 22 (rather than age 18) using household assistance records. The estimated effects are 2.33 percentage points for high school graduation by age 22 ($p = 0.102$), 2.96 percentage points for any college enrollment by age 22 ($p = 0.041$), and 2.67 percentage points for four-year enrollment by age 22 ($p = 0.039$). Measuring treatment through age 24 yields a 2.38 percentage point effect on college graduation by age 24 ($p = 0.048$).

7 Cost-benefit analysis

My results show that housing assistance programs increase children’s educational attainment, reduce their criminal justice involvement, and reduce housing instability. However, housing assistance is costly to provide. In this section, I evaluate the costs and benefits of providing one year of housing assistance to a household with children using the marginal value of public funds (MVPF) framework from [Hendren and Sprung-Keyser 2020](#). The MVPF equals recipients’ willingness-to-pay divided by net program costs:

$$\text{MVPF} = \frac{\Delta W}{\Delta E - \Delta C},$$

where ΔW represents recipients’ willingness-to-pay for one year of housing assistance, ΔE measures the upfront government cost of providing that year of assistance, and ΔC measures reductions in government expenditures or increases in government revenue generated by that one-year assistance spell. I compute each term using estimates from my quasi-experimental design and external estimates from [Reeder 1985](#), [McCollister et al. 2010](#), [Zimmerman 2014](#), and [Hendren and Sprung-Keyser 2020](#). All dollar amounts in this section are in 2023 dollars unless otherwise noted.

7.1 Policy unit and direct costs

The policy unit is one household with children assisted for one year. Elm City Communities’ average annual subsidy in 2023 was \$6,496.36 per assisted person. I estimate that complier households contain 3.459 people and 1.607 children. The direct government cost is therefore

$$G^{hh} = 3.459 \times \$6,496.36 = \$22,472.32.$$

7.2 Fiscal effects and benefits

College enrollment One additional assisted year through age 18 increases four-year college enrollment by age 22 by 3.92 percentage points ($p = 0.017$). To value this effect, I follow [Hendren and Sprung-Keyser \(2020\)](#)’s extrapolation from [Zimmerman \(2014\)](#). Their calibration assigns an induced FIU admission \$142,757 in lifetime pre-tax earnings, \$2,617 in public college costs, and \$2,851 in private tuition in 2012 dollars. The admissions cutoff studied in [Zimmerman \(2014\)](#) increases FIU admission by 23.4 percentage points but state university attendance by 11.9 percentage points. Because my outcome is four-year college enrollment rather than admission, I multiply all three components of college benefits and costs by $0.234/0.119 = 1.966$. After CPI conversion, the resulting values per induced state-university attendee are \$372,547 in earnings, \$6,829 in public costs, and \$7,440 in private tuition.

The dose-response estimate of the effect of housing assistance on four-year college enrollment implies

$$1.607 \times 0.03924 = 0.06308$$

additional four-year enrollees per assisted household-year. This generates \$23,500.87 in projected lifetime earnings, \$4,700.17 in tax revenue under a 20% effective tax rate, and \$430.81 in public college costs. Projected after tax child earnings net of private tuition equal

$$B^{child} = (1 - 0.20) \times \$23,500.87 - \$469.34 = \$18,331.36.$$

Homelessness Among children ages 6–11, one additional assisted year reduces the annual probability of any school-recorded homelessness by 0.765 percentage points ($p < 0.001$). The effect on shelter homelessness, the component with direct public costs, is a 0.181 percentage point reduction ($p = 0.099$). Connecticut HMIS records indicate that family shelter stays last 148.6 days on average. Applying an estimated \$5,552.50 public cost per individual stay and the 3.459 complier-equivalent household members gives

$$H = 3.459 \times 0.001809 \times \$5,552.50 = \$35.71$$

in expected shelter savings per household-year.

Crime In the Connecticut-wide sample, one additional assisted year through age 18 reduces the probability of a felony conviction by age 22 by 1.34 percentage points ($p = 0.097$). I separate victim benefits from government savings. Following [McCollister et al. \(2010\)](#), I construct a nonhomicide victim-cost measure using the national felony conviction mix, excluding criminal justice costs, offender career costs, and homicide risk. I

assume one avoided offense and one potential custody episode per child whose conviction is avoided. I then discount victim harm and custody costs ten years to the year of assistance. The resulting prices are \$8,193.62 in victim harm and \$2,704.10 in marginal custody cost per affected child.¹⁹ Crime reduction contributes

$$\begin{aligned} V &= 0.02160 \times \$8,193.62 = \$177.01 && \text{in victim benefits,} \\ J &= 0.02160 \times \$2,704.10 = \$58.42 && \text{in government savings.} \end{aligned}$$

Victim benefits enter willingness to pay, while custody savings reduce net government cost.

Combining the direct cost and the three downstream fiscal channels gives

$$\begin{aligned} \Delta E - \Delta C &= G^{hh} + P - T - H - J \\ &= \$22,472.32 + \$430.81 - \$4,700.17 - \$35.71 - \$58.42 \\ &= \$18,108.83. \end{aligned}$$

7.3 Willingness to pay and MVPF

I parameterize willingness to pay as

$$\Delta W_\lambda(\theta) = \theta G^{hh} + \lambda B^{child} + V.$$

The parameter θ is the cash-equivalent value of one dollar of housing assistance to the recipient household. I report $\theta = 1$, the dollar-for-dollar benchmark, and $\theta = 0.83$, the estimate in [Reeder 1985](#). The parameter λ governs whether children's after-tax earnings net of tuition enter willingness to pay. If parents fully internalize children's future earnings and can purchase their optimal housing bundle, an envelope theorem argument favors $\lambda = 0$ ([Humphries et al. 2024](#)). If housing market constraints or incomplete financial markets prevent this, $\lambda = 1$ is appropriate. Victim benefits enter both specifications because they accrue outside the recipient household.

Table 10 reports the resulting estimates. Under the dollar-for-dollar transfer benchmark, the MVPF is 1.25 when child earnings are excluded from willingness to pay and 2.26 when they are included. Under $\theta = 0.83$, the corresponding estimates are 1.04 and 2.05.

¹⁹The conviction mix comes from the Bureau of Justice Statistics' [Felony Sentences in State Courts, 2006](#). The custody calculation uses national sentencing shares, a 3% discount rate, and Connecticut's 2024 marginal custody cost, converted to 2023 dollars.

Table 10: MVPF of one household-year of housing assistance

Transfer valuation θ	Transfer and victim WTP $\lambda = 0$	Transfer, victim, and child WTP $\lambda = 1$
0.83	1.04 [0.87, 1.29]	2.05 [1.02, 3.58]
1.00	1.25 [1.05, 1.55]	2.26 [1.20, 3.84]

Notes: Brackets report 95% bootstrap confidence intervals simulation intervals from one million independent normal draws of the four-year college, Connecticut-wide felony conviction, and pooled ages 6–11 shelter effects.

In Section 6, I showed that housing vouchers have larger effects on postsecondary attainment than public housing. Incorporating these differences, as well as program-specific estimates on crime and sheltered homelessness, into program-specific MVPF estimates, I find larger returns to public investment in voucher programs. I summarize the results in Table 11.

Table 11: Voucher- and unit-specific MVPFs

Program	$\theta = 0.83$		$\theta = 1.00$	
	$\lambda = 0$	$\lambda = 1$	$\lambda = 0$	$\lambda = 1$
Voucher	1.01 [0.87, 1.21]	1.89 [1.04, 3.07]	1.22 [1.05, 1.45]	2.10 [1.22, 3.32]
Unit	0.87 [0.75, 1.04]	1.01 [0.26, 2.06]	1.05 [0.90, 1.25]	1.19 [0.41, 2.27]

Notes: Brackets report 95% bootstrap confidence intervals simulation intervals from one million independent normal draws of the four-year college, Connecticut-wide felony conviction, and pooled ages 6–11 shelter effects.

Comparison with existing estimates My results place the returns to New Haven housing assistance in the middle of existing MVPF estimates in the public and urban economics literature. The most conservative specification in Table 10, which uses $\theta = 0.83$ and excludes children’s future earnings from willingness-to-pay, yields an MVPF of 1.04. This is above the baseline housing voucher estimate in [Hendren and Sprung-Keyser 2020](#), which uses results from the Chicago voucher lottery studied by [Jacob et al. 2015](#). In that setting, the implied MVPF is 0.65, rising only to 0.68 when child test score effects are used to forecast future earnings. The lower Chicago estimate reflects the combination of mod-

est estimated child effects and negative fiscal externalities from reduced adult earnings and increased transfer receipt. My direct transfer estimates are larger because I find no evidence of adult labor supply reductions, and because reduced shelter use and incarceration and children's projected future tax payments partly offset the upfront subsidy cost.

The MVPF estimates that include child earnings in willingness-to-pay, 2.05 under the [Reeder 1985](#) valuation and 2.26 under dollar-for-dollar transfer valuation, are larger than recent estimates for housing assistance programs based on their insurance value alone. [Grasley 2026](#) estimates that federal rental assistance has an MVPF of 1.51 once rental assistance is modeled as insurance against uninsurable income risk. In my calculations, the additional value comes from realized human capital effects, measured through increased four-year college enrollment, and from public savings due to reduced shelter use and incarceration.

The estimates are also close to [Chaudhry and Eng 2024](#), who study childhood exposure to New York City public housing. They estimate that each dollar of government spending on public housing generates about \$1.40 in benefits for children overall, and about \$2.30 for children from the most disadvantaged families. My estimates are slightly above their overall estimate but below their estimate for the most disadvantaged children. This comparison suggests that the New Haven effects are meaningful, but not unusually large relative to recent evidence on childhood exposure to housing assistance.

My results should also be interpreted in light of evidence that the effects of housing assistance depend on how these transfers change consumption. [Collinson and Gibbs 2025](#) study housing assistance lotteries in Miami-Dade county and find that vouchers improve early literacy, kindergarten readiness, behavioral development, and even elementary school test scores, while public housing has no detectable average effect on children. They also show that these differences are driven by differential effects on housing and neighborhood quality: vouchers move families to higher-quality housing and safer neighborhoods, while public housing moves families, on average, to lower-quality neighborhoods.

A final benchmark to consider is cash. In principle, a cash transfer allows households to optimally choose consumption, which may be preferable from a welfare perspective. And recent evidence from [Krause et al. 2025](#) shows that large temporary cash transfers increase short-run consumption, including spending on rent, and increase neighborhood mobility. However, the same study finds little persistent improvement in household finances from the transfer. In other words, a low or moderate MVPF from housing vouchers and public

housing does not by itself imply that housing assistance is undesirable if policymakers are paternalistic and cash transfers fail to deliver spillovers to other socially-valuable outcomes (Currie and Gahvari 2008).

8 Conclusion

This paper presents new quasi-experimental evidence on the impacts of housing assistance on children from low-income households. To estimate the effects of these transfers, I construct a new administrative panel dataset spanning nearly two decades that links applications to housing assistance wait lists in New Haven, Connecticut, to records on housing assistance participation, housing instability, children’s educational outcomes, and adult labor market outcomes. I first show that housing assistance significantly reduces school-reported housing instability. I then show that receiving assistance does not improve end-of-grade test scores in elementary and middle school but does raise early childhood development among boys. Next, I show that housing assistance raises rates of high school graduation and college enrollment, with the results driven by children in households that receive tenant-based vouchers. Finally, I calculate the benefit per net dollar spent on federal housing assistance. I find that housing assistance produces large returns through increases in predicted earnings among children in families that lease up after assistance lotteries. My MVPF estimates are close to two, with larger estimates for housing vouchers.

While the setting I study is similar along observable dimensions to many other medium-sized American cities, my results apply to a particular set of applicants and study period. As highlighted by evidence from voucher design, cash transfer, and mobility experiments (Collinson and Ganong 2018; Bergman et al. 2024; Phillips and Sullivan 2025), a promising area for future work is to compare the relative return to short or long-term housing assistance that stabilizes families and housing policies that change children’s exposure to high opportunity neighborhoods. Additional studies of housing assistance, using information on wait list applicants from different settings and with more detailed measurements of non-cognitive skills, other long-run outcomes, or information on parental investments in children, are especially welcome.

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A Appendix Tables and Figures

A.1 Appendix Tables

Table A.1: Child characteristics across lotteries

Lottery	Children	Offered by $t + 5$	Assisted by $t + 5$	Age	Female	Black	Adult Employed
HCV 2003	730	0.94	0.32	5.66	0.48	0.46	0.73
HCV 2005	602	0.73	0.20	5.93	0.50	0.56	0.77
LIPH 2007	1,271	0.74	0.24	4.74	0.50	0.59	0.74
LIPH 2008	575	0.48	0.20	4.38	0.46	0.56	0.77
HCV 2011	1,702	0.40	0.19	7.55	0.49	0.47	0.68
LIPH 2012	549	0.31	0.19	3.85	0.50	0.78	0.71
LIPH 2013	671	0.40	0.19	4.57	0.50	0.66	0.69
HCV 2015	132	0.36	0.23	6.83	0.48	0.60	0.90
HCV 2018 [†]	1,167	0.31	0.07	5.86	0.49	0.64	0.83

Notes: Table A.1 reports characteristics of children at the level of the lottery. The offered and assisted columns report the share of children in households selected from the wait list (offered) or receiving assistance (assisted) by five years after the lottery year, the final year available for all lotteries in the sample. Age is the child's age at the time of the lottery. Female and Black are means of child demographic indicators. Adult employed is the share of children in families with any adult observed employed in the year before application, calculated among children in the linked SDE & DOL sample. The data source is the restricted lottery sample, which excludes child-lottery observations whose household received housing assistance before the lottery year. The HCV 2018 lottery is denoted with [†] because it is a "project based voucher" lottery.

Table A.2: Offer timing and linkage to administrative records

	(1) Raw	(2) Regression adjusted
SDE linked	−3.77*** (1.24)	1.35 (1.30)
SDE and DOL linked	−4.12*** (1.26)	1.36 (1.33)
Lottery fixed effects	No	Yes
Age fixed effects	No	Yes
Lottery × family size × disability FE	No	Yes
Demographic controls	No	Yes
SDE linkage rate (%)	75.55	75.55
SDE and DOL linkage rate (%)	74.40	74.40
Offered by year five (%)	52.32	52.32
Children	7,399	7,399
Households	4,194	4,194

Notes: This table reports OLS estimates of the impact of receiving an offer by five years after a lottery on administrative data linkage. Coefficients are reported in percentage points. Column (1) is unadjusted. Column (2) includes age fixed effects, fully interacted lottery × family size × household head disability fixed effects, and controls for race, sex, and citizenship. Standard errors, shown in parentheses, are clustered by household. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$, respectively.

Table A.3: Effects of housing assistance on latent developmental factor by gender

	Estimate	Bootstrap SE	p -value
<i>Panel A: Binary 2SLS effects on the developmental factor</i>			
Pooled	0.14	0.12	0.230
Boys	0.35**	0.17	0.040
Girls	0.01	0.17	0.930
Difference (Boys – girls)	−0.34	—	0.156
N		2,226	

Notes: Panel A reports binary 2SLS effects in standard deviations of the latent developmental readiness factor. Creativity, physical/motor development, and personal/social development load on this factor. The endogenous treatment is assistance receipt by the kindergarten assessment year, instrumented with an offer by that year. Bootstrap inference resamples households and re-estimates the measurement and causal models. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$, respectively.

Table A.4: Effects of housing assistance on test scores, by grade and gender

Grade	Pooled		Boys		Girls		Girls – boys
	2SLS	Children	2SLS	Children	2SLS	Children	Difference
<i>Panel A: Math</i>							
Grade 3	0.145 (0.130)	1,440	0.246 (0.202)	693	0.088 (0.182)	747	-0.158 (0.269)
Grade 4	0.059 (0.114)	1,783	0.115 (0.175)	907	-0.013 (0.158)	875	-0.128 (0.234)
Grade 5	0.039 (0.105)	1,914	0.184 (0.158)	965	-0.076 (0.147)	948	-0.260 (0.212)
Grade 6	0.142 (0.113)	2,039	0.113 (0.167)	1,025	0.226 (0.159)	1,013	0.112 (0.228)
Grade 7	-0.020 (0.120)	1,948	-0.004 (0.195)	975	0.011 (0.156)	973	0.015 (0.240)
Grade 8	0.023 (0.133)	1,878	0.006 (0.208)	903	0.010 (0.165)	975	0.004 (0.256)
<i>Panel B: English language arts</i>							
Grade 3	0.159 (0.132)	1,445	0.446** (0.214)	694	0.028 (0.176)	751	-0.418 (0.271)
Grade 4	0.067 (0.121)	1,794	0.263 (0.186)	914	-0.109 (0.172)	879	-0.371 (0.252)
Grade 5	0.068 (0.115)	1,913	0.339* (0.175)	966	-0.086 (0.157)	946	-0.425* (0.231)
Grade 6	0.028 (0.117)	2,052	-0.048 (0.174)	1,030	0.159 (0.163)	1,021	0.206 (0.233)
Grade 7	-0.078 (0.122)	1,977	0.000 (0.199)	992	-0.073 (0.160)	985	-0.073 (0.248)
Grade 8	-0.214 (0.143)	1,927	-0.224 (0.232)	926	-0.214 (0.178)	1,001	0.010 (0.285)

Notes: Table A.4 reports 2SLS estimates of the effect of receiving housing assistance by the assessment year on standardized math and English language arts scores. Standard errors, shown in parentheses, are clustered at the household level. First stage F statistics range from 340.5 to 542.4 in the pooled models, from 158.0 to 276.4 for boys, and from 192.7 to 293.1 for girls. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table A.5: Effects of housing assistance on adult employment and earnings, by child age

	Ages 6–7	Ages 8–9	Ages 10–11	Ages 12–13	Ages 14–15	Ages 16–17
<i>Panel A: Effect of receiving assistance by the corresponding child age</i>						
Household adult earnings	2,322 (2,389)	2,488 (2,552)	2,973 (3,169)	3,181 (4,039)	7,036 (4,932)	12,432** (6,236)
IHS household adult earnings	0.175 (0.435)	0.653 (0.473)	1.012* (0.569)	0.771 (0.685)	0.891 (0.822)	1.221 (1.029)
Any adult employed	0.23 (3.94)	5.72 (4.28)	8.41 (5.12)	7.37 (6.15)	8.30 (7.31)	8.52 (9.11)
First stage <i>F</i> -statistic	906.2	874.1	714.7	565.2	408.3	255.6
<i>Panel B: Effect of one additional year of assistance by the corresponding child age</i>						
Household adult earnings	417 (514)	434 (458)	746 (520)	400 (558)	1,065 (717)	2,073** (964)
IHS household adult earnings	0.021 (0.100)	0.067 (0.086)	0.181** (0.091)	0.157* (0.094)	0.120 (0.118)	0.156 (0.151)
Any adult employed	0.06 (0.91)	0.57 (0.78)	1.52* (0.82)	1.62* (0.84)	1.11 (1.04)	1.13 (1.34)
First stage <i>F</i> -statistic	623.3	695.8	585.3	499.2	327.4	210.6
Observations	6,850	7,511	7,672	7,507	6,540	5,474
Children	3,672	3,999	4,095	3,992	3,585	2,971
Households	2,556	2,642	2,598	2,505	2,244	1,844

Notes: Table A.5 reports 2SLS estimates of the effects of housing assistance on employment and earnings among adult household members identified at the child's selected lottery application. Each column pools observations within the indicated age bin. Panel A instruments an indicator for receiving assistance by the corresponding age with an indicator for having received an offer by that year. Panel B instruments cumulative years of assistance with cumulative years of offer exposure. Earnings are measured in 2023 dollars. Employment coefficients and standard errors are reported in percentage points. Standard errors, shown in parentheses, are clustered at the household level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

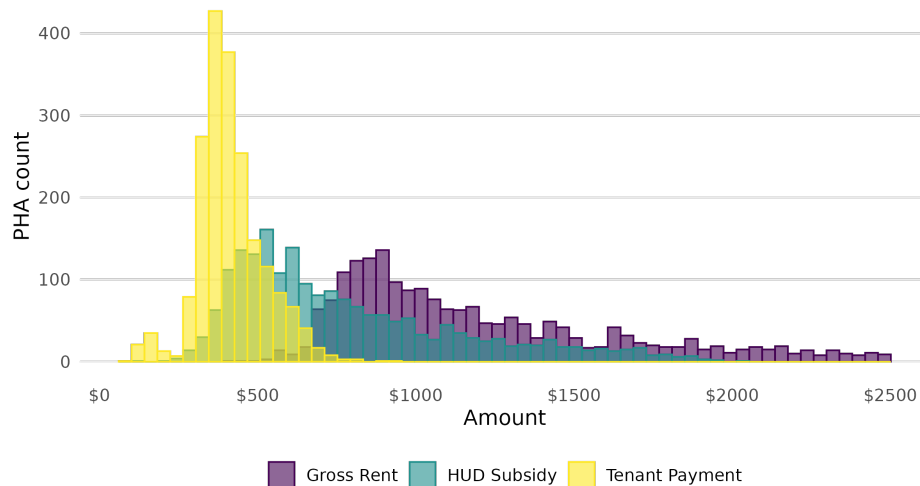
Table A.6: Effects of housing assistance on school quality and composition, by child age

	Ages 6–7	Ages 8–9	Ages 10–11	Ages 12–13	Ages 14–15	Ages 16–17
<i>Panel A: Effect of receiving assistance by the corresponding child age</i>						
ELA value added	-0.007 (0.019)	-0.007 (0.016)	-0.001 (0.013)	0.019 (0.016)	— —	— —
Math value added	-0.007 (0.022)	-0.022 (0.018)	-0.005 (0.013)	0.005 (0.016)	— —	— —
School proficiency rate	0.06 (1.92)	0.09 (1.74)	-1.60 (1.61)	-2.37 (1.85)	— —	— —
School FRPL share	-0.55 (1.54)	-0.74 (1.54)	1.95 (1.57)	3.71* (1.91)	5.92*** (2.21)	7.21** (2.85)
School homeless share	-0.28 (0.23)	-0.14 (0.25)	0.04 (0.23)	0.30 (0.20)	0.42* (0.23)	0.30 (0.31)
First stage <i>F</i> : value added sample	366.8	481.4	569.5	315.9	—	—
First stage <i>F</i> : proficiency sample	467.1	588.7	612.0	436.1	—	—
First stage <i>F</i> : composition sample	749.9	754.1	737.8	584.7	443.3	269.7
<i>Panel B: Effect of one additional year of assistance by the corresponding child age</i>						
ELA value added	0.001 (0.004)	0.000 (0.002)	0.004** (0.002)	0.002 (0.002)	— —	— —
Math value added	-0.003 (0.005)	-0.002 (0.003)	0.003 (0.002)	0.003 (0.002)	— —	— —
School proficiency rate	0.44 (0.39)	0.15 (0.28)	0.26 (0.23)	0.06 (0.23)	0.16 (0.41)	-0.46 (0.57)
School FRPL share	-0.28 (0.31)	-0.27 (0.26)	0.04 (0.23)	0.38 (0.25)	0.55** (0.27)	0.99*** (0.37)
School homeless share	-0.11** (0.05)	-0.09** (0.04)	-0.05 (0.03)	-0.02 (0.03)	0.00 (0.03)	0.03 (0.04)
First stage <i>F</i> : value added sample	262.8	378.0	435.1	320.7	—	—
First stage <i>F</i> : proficiency sample	368.4	469.9	516.4	404.4	—	—
First stage <i>F</i> : composition sample	542.5	584.4	615.5	531.1	394.1	228.7
Value added observations	1,977	2,916	3,344	2,560	—	—
Proficiency observations	2,910	3,864	4,222	3,483	—	—
Composition observations	5,031	5,760	6,110	5,853	5,031	3,149

Notes: Table A.6 reports 2SLS estimates of the effects of housing assistance on the leave-one-out characteristics of children's schools. Each column pools observations within the indicated age bin. Panel A instruments an indicator for receiving assistance by the corresponding age with an indicator for having received an offer by that year. Panel B instruments cumulative years of assistance with cumulative years of offer exposure. Standard errors, shown in parentheses, are clustered at the household level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

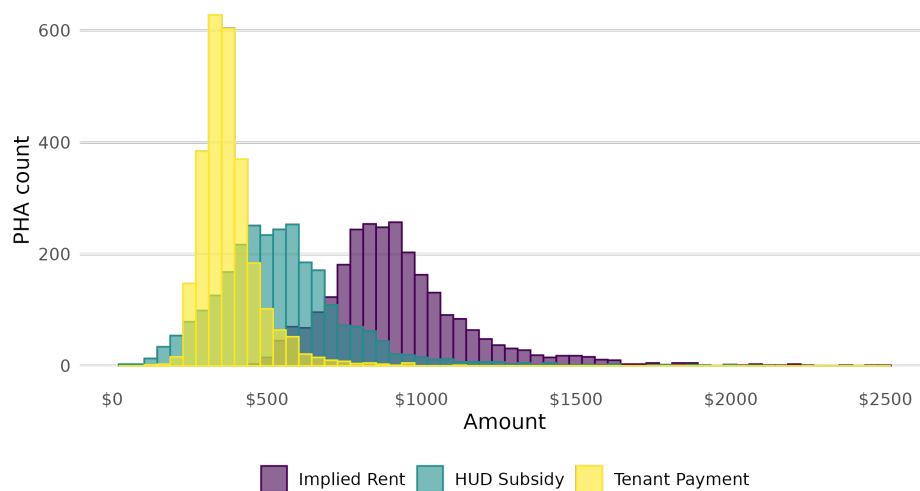
A.2 Appendix Figures

Figure A.1: Tenant payments, gross rents, and **voucher** subsidies across PHAs



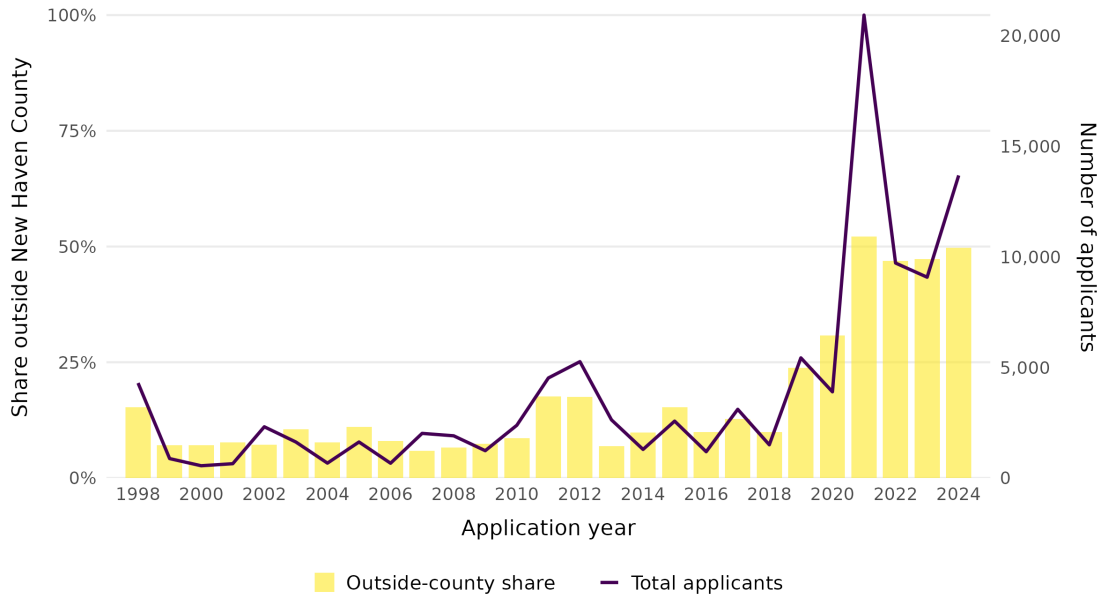
Note: Figure A.1 plots the distribution of monthly tenant payments, gross contract rents, and PHA expenditures (HUD subsidies) in the federal Housing Choice Voucher program in 2023. All dollar values are in 2023 dollars. The results are shown in \$40 bins. HUD records contain 2,024 PHAs with non-missing housing voucher data for 2023. The New Haven PHA has an average subsidy of \$1063.

Figure A.2: Tenant payments, implied rents, and **public housing** subsidies across PHAs



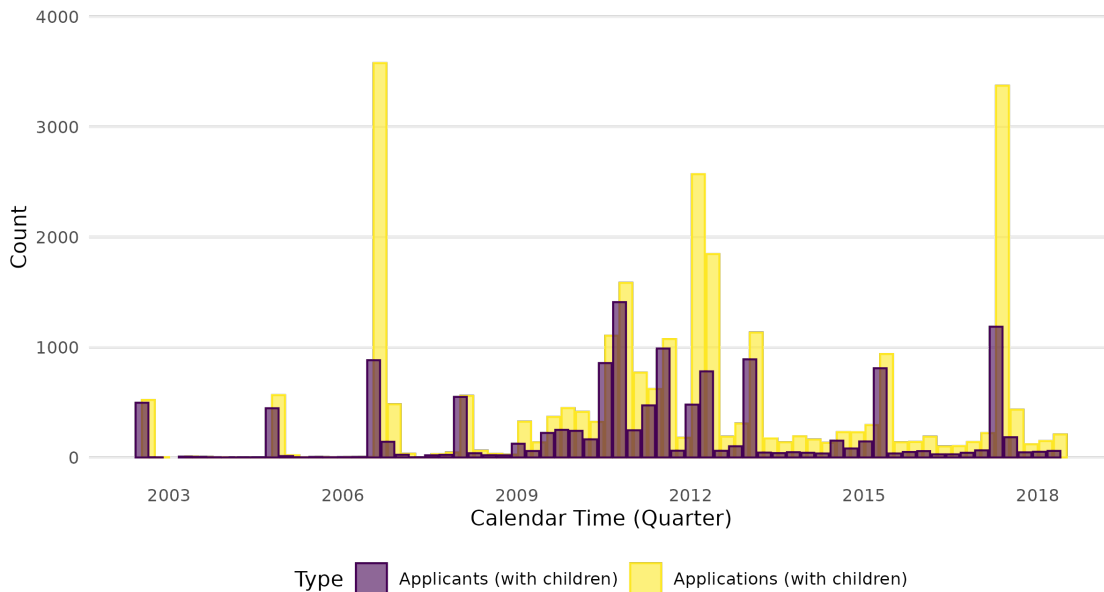
Note: Figure A.2 plots the distribution of monthly tenant payments, implied rents, and PHA expenditures (HUD subsidies) in the federal Low Income Public Housing program in 2023. The results are shown in \$40 bins. All dollar values are in 2023 dollars. HUD records contain 2,646 PHAs with non-missing public housing data for 2023. The New Haven PHA has an average subsidy of \$2068.

Figure A.3: Time series of ECC applicant counts and outside county share



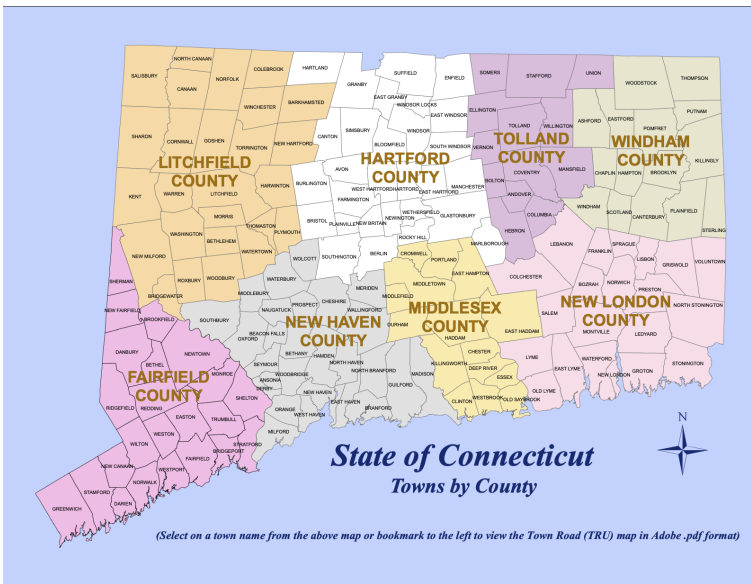
Note: The left y-axis in Figure A.3 is the share of applicants from outside of New Haven county, measured using the last recorded ZIP per applicant. The right y-axis is the number of applicants. Each series is recorded annually from 1998-2024. The data source is the universe of ECC applications.

Figure A.4: ECC wait list applications from households with children



Note: The y-axis is the count of applications (applicants) to one of the 24 Elm City Communities wait lists in the descriptive sample, limited to households with children. The x-axis is the calendar time in quarters from January 2003 to December 2018. The data source is the descriptive sample of ECC applications, limited to households with children.

Figure A.5: Map of Connecticut counties



Note: Figure A.5 displays a map of Connecticut counties. New Haven county is shaded in grey. The matched SDE sample contains all Connecticut students ever enrolled in a public school in New Haven county during the 2010-2011 to 2023-2024 school years.

PUBLIC NOTICE

ROCKVIEW APARTMENTS

PRE-APPLICATIONS for 30 new Public Housing units will ONLY be available at:

ROCKVIEW LEASING OFFICE located at:

6 Solomon Crossing; New Haven, CT, 06515;

Pre-Applications will be available for the NEXT 90 DAYS;

BEGINNING TUESDAY May 28th, 2013

Applications will no longer be available after Monday AUGUST 26th, 2013

Applicants already on waiting lists for other housing programs must apply separately for Rockview, and such applicant will not lose their place on other waiting lists when they apply for public housing.

Office Hours Monday-Friday 9am- 5pm

Saturday by Appointment Only

All Pre-Applications will be placed on the waiting using a – Lottery.

ONLY ONE PRE-APPLICATION PER HOUSEHOLD WILL BE ACCEPTED.

If you are physically unable to pick up an application,
CALL 203.691.1332 TTY: 711 and one will be mailed out to you.

Rent is based on 30% of Monthly Adjusted Income:			
Unit Size	Number of Public Housing Units	MAXIMUM INCOME RESTRICTIONS	
2 bedrooms	21	2 Persons	\$16,525
3 bedrooms	8	3 Persons	\$18,600
4 bedrooms	1	4 Persons	\$20,650
		5 Persons	\$22,700
		6 Persons	\$23,800
		7 Persons	\$25,000
		8 Persons	\$27,200

PREFERENCES FOR PUBLIC HOUSING APARTMENTS APPLY AS FOLLOWS:

- a. First preference – all residents of Rockview at the time of the execution of the MOA between TRCs of West Rock and HANH.
- b. Second preference— all residents of Brookside at the time of the execution of the Memorandum of Agreement (MOS) (7/17/1999) between TRCs of West Rock and HANH by order of when people initially moved into the development
- c. Third preference—all residents of Westville Manor or Ribicoff Cottages at the time of the execution of the MOA between TRCs of West Rock and HANH.
- d. Fourth preference—applicants in accordance with all other preference set forth in the ACOP.

A priority is given for occupancy of the handicapped accessible units to eligible households who require the special handicapped-accessible features of those units. *A CARES Program Community

Professionally Managed by



*Restrictions Apply. If you have any questions, please call 203.691.1332




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Figure A.7: Housing voucher wait list application form

HCVP / 2021

FOR OFFICE USE ONLY Application Entered By: _____ Application Entered On: _____ Elderly/Disabled Housing _____ General Developments _____ Bedrooms 0 1 2 3 4 5 6	DATE AND TIME STAMP
---	--

HOUSING AUTHORITY OF THE CITY OF NEW HAVEN
360 Orange Street
New Haven, Connecticut 06509-1912
 475-355-7289 (Voice) or (203) 497-7901 (Fax)
 (203) 497-8434 (TDD)

Pre-Application for Housing Choice Voucher Program - Section 8
Equal Housing Opportunity

This is not the full application form for the Housing Choice Voucher (Section 8) waitlist. The information which you are being asked to provide as the head of household is used to determine if your household appears to be eligible to be added to the Housing Choice Voucher (Section 8) waitlist. You will be required to complete a Full Application prior to any final processing for an offer of a voucher. All information is subject to third party verification, and you will be required to sign releases that will permit the Housing Choice Voucher Program to verify all information provided below. By signing this application, you are certifying that the information you have provided is correct and that your household is within the income limits for the program as of the date of signature. **Misrepresentation of information is grounds for immediate removal from the waiting list or termination from the Conventional Public Housing Programs.**

For applicants to federal housing, Title 18, Section 1001 of the U.S. Code states that it is a felony to intentionally make false or fraudulent statements to any federal department or agency. As the information provided below may be shared with the U.S. Department of Housing and Urban Development, misrepresentation of information is a felony.

Incomplete Pre-applications will not be processed. It is the responsibility of the applicant to provide all required information and answer all questions completely. All applications are the property of the Housing Authority of the City of New Haven.

Assistance Available: For general questions call 475-355-7289 or TDD 203-497-8434. If you need assistance completing this application, please see the receptionist or call (203) 498-8800 ext. 1507 for an appointment.

Please print all Answers in a Legible Fashion

1. Head(s) of Household: _____

2. Residential Address: _____

City or Town _____ State _____ Zip Code _____

3. Current Mailing Address: _____

City or Town _____ State _____ Zip Code _____

4. Email Address: _____

5. Home Phone (____) _____ Work Phone (____) _____

02/25/2021

1

Note: Figure A.7 presents the pre-application form for the Elm City Communities HCV wait list. Form pages two and three are shown on the following pages.

5. Is there a member of your household who requires a physically modified unit or an exception to our policies in order to address a disability?

☐ YES ☐ NO

If so, please list which household member(s) with their first and last name?

PLEASE INDICATE THE ACCOMMODATION NEEDED BY PLACING A CHECKMARK IN AS MANY OF THE BOXES BELOW THAT APPLY TO YOUR HOUSEHOLD:

- | | |
|--|--|
| ☐ Barrier-free unit (wheelchair accessible) | ☐ Bathroom and bedroom on 1 st floor |
| ☐ Unit adapted for vision impairments | ☐ Counter spaces in kitchen /bathroom lowered |
| ☐ Unit adapted for hearing impairments | ☐ Live-in-Aide |
| ☐ Ramp/Elevator in Building | ☐ Parking space close to unit |
| ☐ Other (please list): | |

6. Please provide the full name including middle initial of all household members, their date of birth, place of birth, sex, relationship to the head of household, and **Social Security Number or attach proof of application for a Social Security Number**. If any of this information is not provided, the pre-application will be considered incomplete and will be rejected.

***Race and Ethnicity are optional.**

HUD's race codes are: White, Black, American Indian/Alaskan Native, Asian, and Native Hawaiian/Other Pacific Islander. HUD's ethnicity codes are: Hispanic, or Not Hispanic. Please use the HUD race and ethnicity codes that best describe *each* member of your family. For example: White/Hispanic, or Black/Non-Hispanic, etc.

Only the race/ethnicity column is optional.

Name	Date of Birth	Place of Birth	Sex	Relation to Head of Household	Social Security Number	Race/Ethnicity : *Optional
				Head	- -	
					- -	
					- -	
					- -	
					- -	
					- -	
					- -	
					- -	
					- -	
					- -	
					- -	

Figure A.7: (continued)

7. Current Household Annual Income for all sources: \$ _____.
(A determination of income, assets and deductions will be made as part of the final application review.)

Please complete this section based on ALL income/money coming into the household for ALL family members.

Family Member	Type of Income (Employment, Welfare, SSI, Child Support, etc)	Amount received	Weekly, Bi-weekly, Monthly, or Annually	Source of Income (Public Assistance, Name of Employer or Company, etc.)

8. Would you like us to direct all communications regarding your application to you **OR** to an emergency contact/caseworker/other? **Please Check Only One:**

- ☐ Please send all future communications regarding this application to me.
- ☐ Please send all future communications to my emergency contact /caseworker or other person.

Emergency Contact / Caseworker / Other _____
Relationship/Organization _____
Mailing Address _____
City _____ State _____ Zip Code _____ Phone # (____) _____

I understand that this pre-application is not an offer of an apartment. I certify that my household is income eligible under current program income limits and the information contained in this application is true and complete under pains and penalty of perjury. I agree to authorize the Housing Authority to make inquiries to verify the information I have provided on this application. I understand that it is my responsibility to inform the Housing Authority of any change in address or in household composition, in writing.

Applicant's Signature

Date

Co- Applicant Signature

Date

Figure A.7: (continued)

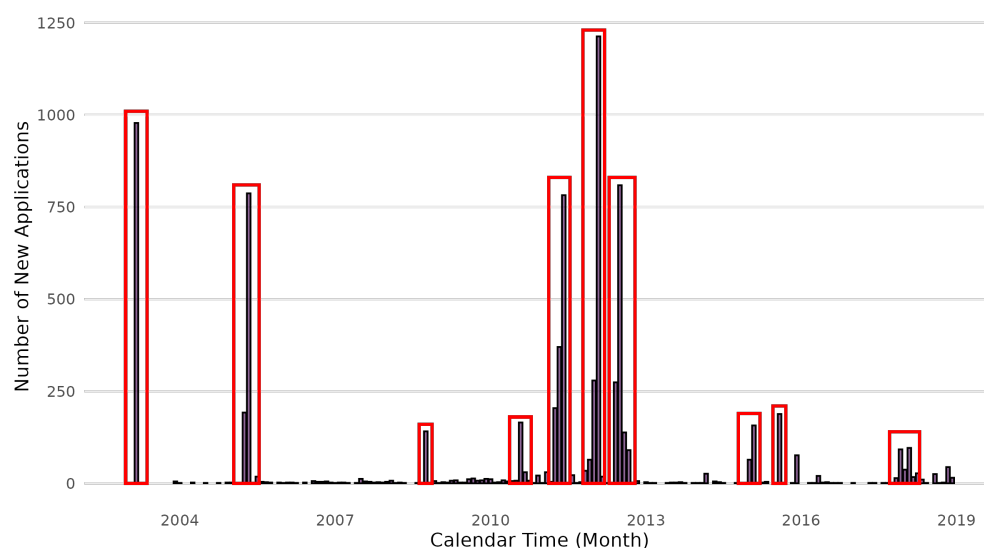
B Classifying wait list opening periods and lotteries

Wait lists for housing assistance are always open to households that are displaced by government action, include victims of domestic violence or sexual assault, are in a documented Witness Protection Program, or have direct referrals from the homelessness Coordinated Access Network. Households that do not meet these conditions may only apply for housing assistance from a particular PHA when its wait lists are open.

Applicants from outside traditional opening periods likely differ along observable and unobservable dimensions from other applicants. To address this concern, I identify wait list opening events by manually isolating structural breaks in the wait list-by-month count of new applications and cross referencing these structural breaks with local news articles announcing wait list openings. I use these wait list opening periods to define a subset of the descriptive sample that uses only clearly-defined wait list openings. I next combine these wait list openings with information on the assigned wait list position of applicants. To identify lotteries, I require that an applicant's wait list position be uncorrelated with the exact date and time of their application *within* the wait list opening period.

Opening periods In Figure B.1, I illustrate wait list opening periods on the tenant-based HCV wait list from 2003-2018.

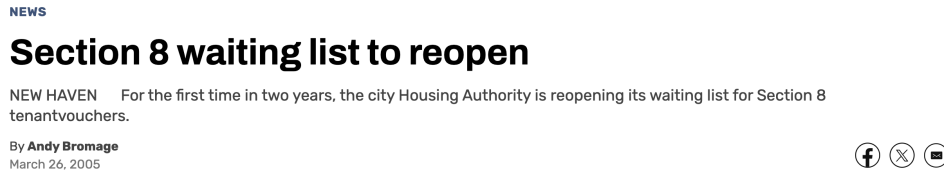
Figure B.1: HCV wait List Applications



Note: The y-axis is the count of new applications to Elm City Communities' tenant-based HCV wait list. The x-axis is the calendar time in months from January 2003 to December 2018. The red rectangles highlight wait list opening periods. Data comes from ECC wait list application records.

I identify 10 wait list opening periods: March 2003, April-May 2005, October 2008, August 2010, April-June 2011, January-February 2012, June-September 2012, January-February 2015, August 2015, and December 2017-February 2018. Whenever possible, I corroborate the identified openings using local news announcements or announcements of wait list opening events in ECC reports submitted to HUD annually from FY 2006 to FY 2023 (excluding FY 2007). For example, below is the header for a March 2005 article in the *New Haven Register* announcing the reopening of the HCV wait list:

Figure B.2: New Haven Register wait list announcement



The body of the article - available [here](#) as of June 2026 - describes a lottery for 1000 spots on the wait list and notes that the wait list last opened in March 2003. I document 945 successful tenant-based HCV wait list applications in April and May 2005 during the lottery period.

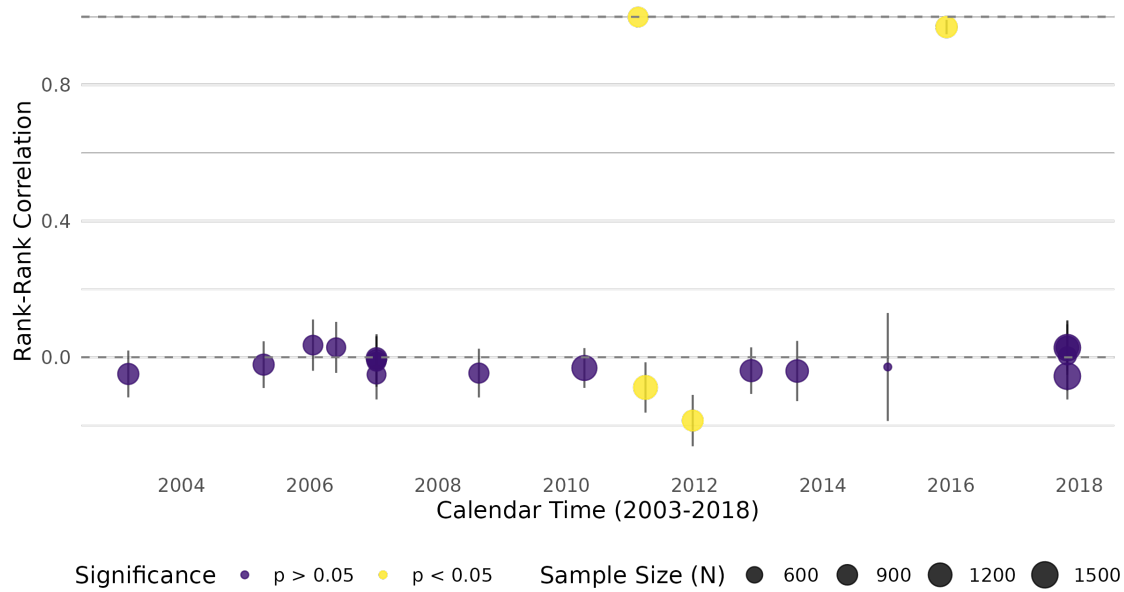
Lotteries To determine whether wait list opening events are lotteries, I check that the lottery ordering is not equivalent to ordering households by the exact date and time of application. Separately for each wait list opening j , I estimate – using the subsample of applicants without overwritten positions – the correlation β_j between the system-assigned position R_{ij} and the position an applicant would receive given the exact date and time of their application R_{ij}^{Date}

$$R_{ij} = \alpha_j + \beta_j R_{ij}^{\text{Date}} + \epsilon_{ij}. \quad (4)$$

I identify an opening period as a lottery whenever I fail to reject the null hypothesis that $\beta_j = 0$ at the 5% significance level. Some applications have lottery position that are overwritten at the time of interview offer or when the household re-applies in wait list openings after 2018. In Figure B.3, I graphically present the coefficient estimates and standard errors from separately estimating Equation (4) for a subset of the manually-identified wait list opening periods from 2003-2018 across all wait lists²⁰. The sample sizes are for the entire set of applicants regardless of overwritten position.

²⁰I remove wait list openings restricted to elderly and disabled applicants. For example, the October 2008 HCV wait list opening is restricted to elderly and disabled applicants

Figure B.3: System position - date position correlation



Note: The y-axis is the $\hat{\beta}_j$ point estimate and 95% confidence interval from estimating Equation 4. The upper dashed line is at 1 where date position and system position are identical. The lower dashed line is at 0 where date position and system position are unrelated. The significance level is for a hypothesis test of $\beta_j = 0$. Data comes from ECC wait list application records.

It is apparent from examining the $\hat{\beta}_j$ point estimates that most well-defined opening periods from 2003-2018 are lotteries. The regression test identifies 9 lottery opening periods across 14 wait lists and 4 non-lottery opening periods.

C Data appendix

C.1 Education data

State employees match ECC children to State Department of Education (SDE) records using first name, last name, date of birth, and the last 4 digits of the Social Security number. They match 5,590 children in the descriptive sample to unique SDE records (75.6%). Of these children, 5,505 are in households with a unique adult DOL match (98.4%).

C.1.1 Kindergarten test scores

This section describes my measures of student achievement for kindergarteners, which are the earliest measures of achievement observed in the SDE records. This section closely follows [Humphries et al. 2024](#), which uses identical achievement measures to study the effects of universal pre-kindergarten (UPK) programs in New Haven.

I use the Connecticut Kindergarten Entrance Inventory (KEI) to measure student achievement in kindergarten. Kindergarten teachers conduct this assessment in the fall of each academic year. Students receive scores ranging from one to three on six readiness measures: language proficiency, numeracy, literacy, physical/motor skills, creativity, and personal/social skills. A higher score represents both a more consistent demonstration of the skills and that the student requires less instructional support for the skills. I observe KEI scores for each school year from 2008-2009 through 2023-2024.

To construct the measures of kindergarten achievement used in the paper, I standardize each dimension of the KEI within each cohort. For dimension k , the standardized score for student i in year t is given by

$$z_{it}^k = \frac{KEI_{it}^k - \overline{KEI_t^k}}{\sigma_t^k},$$

where the mean and standard deviation are taken over the population of New Haven county kindergarten students. I aggregate these scores into a KEI achievement index

$$\bar{z}_{it} = \frac{\sum_{k=1}^6 z_{it}^k}{6}.$$

C.1.2 End-of-year test scores

This section describes my measures of student achievement for students in grades 3-8, which are the most consistent measures of achievement observed in the SDE records.

I use the Connecticut Smarter Balanced Assessment (SBA) to measure student achievement in grades 3-8. SBA records are available beginning in the 2014-2015 school year. The SBA measures student end-of-year achievement in English Language Arts (ELA) and Mathematics. The SBA consists of a computer adaptive test (in ELA and Mathematics) and a performance task (in Mathematics). Students are assigned a vertical score (which ranges from 2,114-2,862) and a performance level (which ranges from 1-4) in each category. Students assigned a performance level of 1 or 2 do not meet or are approaching expectations. Students assigned a performance level of 3 or 4 meet or exceed expectations. In this paper, I use the vertical score to measure student achievement. I standardize the score within each grade-year for the full sample of New Haven county students so that the ELA and Mathematics scores are given by

$$z_{it}^E = \frac{s_{it}^E - \overline{s_{g(i)t}^E}}{\sigma_{g(i)t}^E} \quad \text{and} \quad z_{it}^M = \frac{s_{it}^M - \overline{s_{g(i)t}^M}}{\sigma_{g(i)t}^M}$$

where $g(i)$ denotes grade of student i and s_{it}^E and s_{it}^M are the vertical score in ELA and Mathematics for student i during year t .

C.1.3 Other outcomes

This section describes the other outcomes that I measure using the SDE records. I focus on student attendance and chronic absenteeism in addition to homelessness and housing instability. These data are available from the 2010-2011 to 2023-2024 school years.

Attendance I observe the number of school days attended by each student, the number of possible school days, and an indicator for chronic absenteeism. In the full sample of New Haven county students, 17.4% of student-years involve chronic absenteeism (i.e., missing more than 10% of instructional days in a school year). The average attendance rate is 93.4%, and 44.6% of students are chronically absent in at least one school year²¹. Among all matched ECC applicants, the average attendance rate is 89.6% and 73.2% of students are chronically absent in at least one school year. Within the matched lottery

²¹The average attendance rate is taken across all student-years. As such, this number is a weighted average across students with the weights in proportion to the number of school years per student.

sample, 79.5% of students are chronically absent in at least one school year. The average attendance rate in the lottery sample is 90.1%.

Homelessness The McKinney-Vento Act of 1987 requires public schools to collect information on student homelessness and housing insecurity. The Act requires school districts to appoint local homeless-education liaisons who identify students experiencing homelessness – often through enrollment questionnaires, staff referrals, and outreach – to connect them with academic and support services. The SDE records identify when students are experiencing one of 4 different types of housing instability: doubling up, living in a shelter, living in a hotel or motel, and experiencing unsheltered homelessness. In the lottery sample, 7.1% of children ever experience doubling up, 4.4% of children are ever recorded in a homeless shelter, and 2.2% of children are ever recorded living in a hotel or motel. Rates of unsheltered (i.e., street) homelessness are very low.

C.1.4 Test score value added

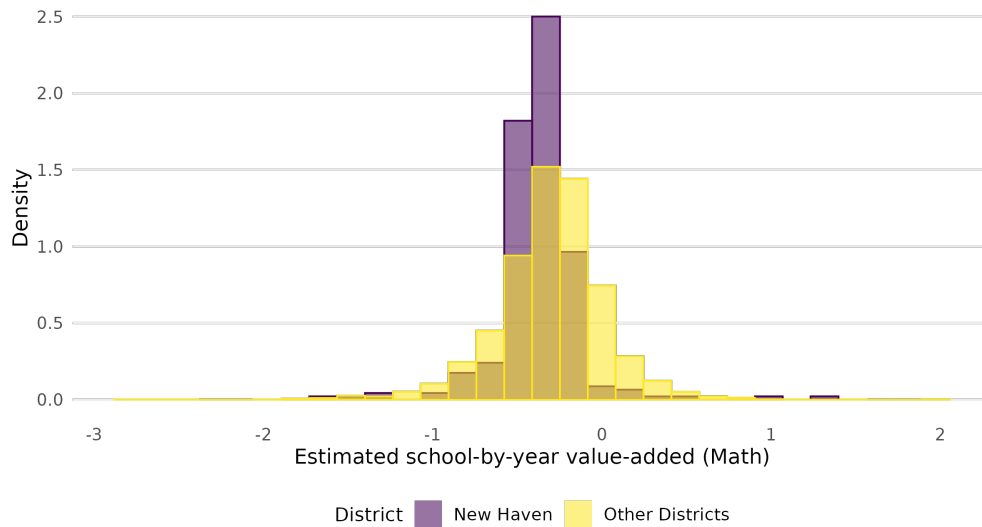
In this section, I present a simple statistical model of test score value added in the spirit of [Hanushek 1971](#), which I use to characterize the quality of schools available to students in New Haven county in grades 3-8.

Let A_{igst} denote student i 's end-of-grade test score in grade g at school s in year t . I standardize scores within each grade-year to obtain Z_{igst} . I estimate school value added from the lagged-score model

$$Z_{igst} = \theta Z_{i,g-1,s',t-1} + \beta X_{it} + \gamma_{st} + \epsilon_{igst}, \quad (5)$$

where X_{it} is a vector of student-level controls, and γ_{st} is the school-by-year value added parameter.²² Because information on teachers is not observed in the state administrative records, the $\hat{\gamma}_{st}$ estimates combine pure “school” effects and the impacts of teacher sorting across schools ([Chetty et al. 2014](#)). I construct the $\hat{\gamma}_{st}$ estimates using OLS. I plot the distribution of school-by-year fixed effects separately for math and ELA in Figure C.1 (math) and Figure C.2 (ELA) below, with separate series for New Haven schools and other schools in New Haven county.

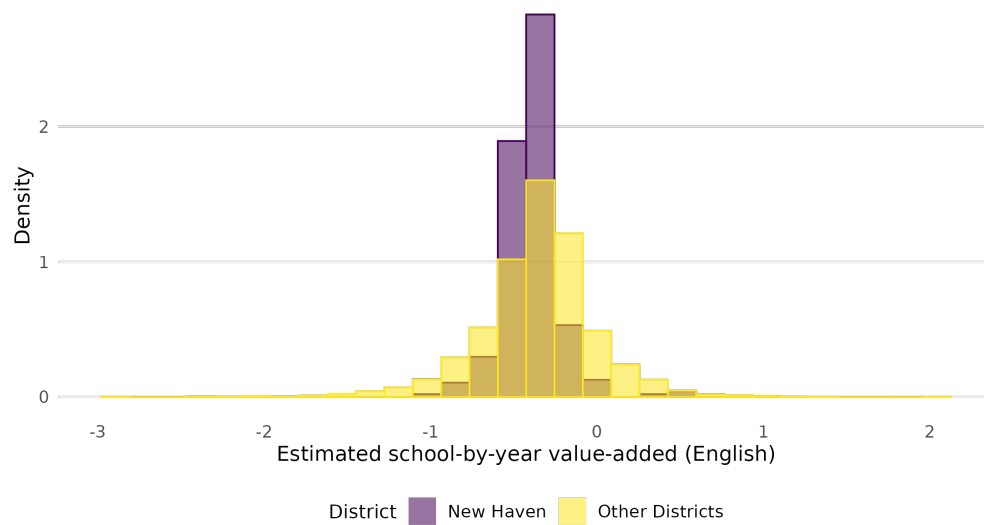
Figure C.1: Grade 3-8 math value added distribution



Note: Figure C.1 displays the distribution of school-by-year fixed effects from estimating Equation 5 using end-of-year math test scores for students in grades 3-8. School-year fixed effects are calculated for the 2015-2016 to 2023-2024 school years.

²²The X_{it} vector contains information on student i 's contemporaneous FRPL status, homelessness status, and gifted and talented program status, as well as lagged attendance.

Figure C.2: Grade 3-8 English value added distribution



Note: Figure C.2 displays the distribution of school-by-year fixed effects from estimating Equation 5 using end-of-year ELA test scores for students in grades 3-8. School-year fixed effects are calculated for the 2015-2016 to 2023-2024 school years.

C.2 Earnings and employment data

In this section, I describe the earnings and employment records provided by the Connecticut Department of Labor (DOL). DOL provides me with a panel of earnings at the individual-by-quarter-by-job level (employer information is not provided), containing all wages earned in Connecticut covered by unemployment insurance.²³ The file includes anonymous individual identifiers, each job's 6-digit NAICS code, and the individual's earnings in each job-quarter with nonzero recorded earnings. The observations run from the first quarter of 2001 to the final quarter of 2021.

DOL attempts to match the full universe of ECC applicants to their earnings records using a combination of name, date of birth, and the last 4 digits of the Social Security Number. DOL successfully assigns a unique matched identifier to 90.6% of all ECC applicants.

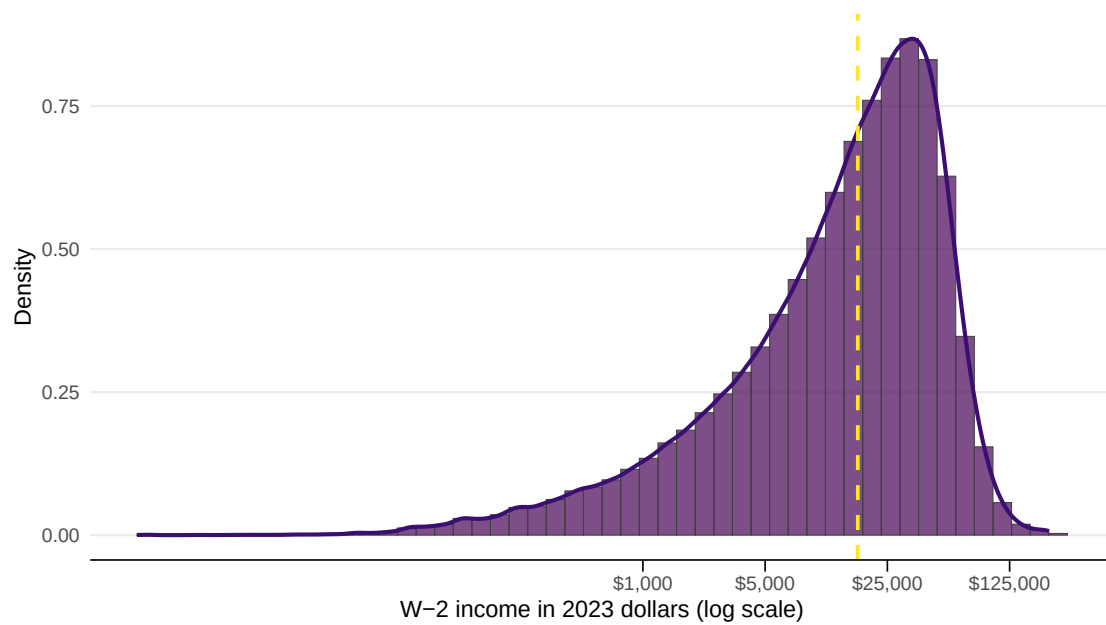
Data processing Starting from the 149,885 uniquely-matched individuals, I construct an individual-by-quarter panel from Q1 2001 - Q4 2021. The value for individual earnings in each quarter is the sum of quarter-by-job earnings for each individual. In periods when an individual is missing from the DOL earnings panel, I assume that the individual's earnings in that quarter are \$0.²⁴ To standardize the earnings measures over time, I deflate quarterly earnings using the January 2023 CPI, meaning that all earnings measures are in 2023 dollars.

I produce a balanced panel of 12,590,340 observations for 149,885 individuals. I top-code individual earnings in the full panel at the 99.9th percentile of quarterly real earnings in the conditional distribution of earnings given employment in the sample (\$52,400.07 in 2023 dollars). Figure C.3 shows the conditional distribution of earnings in the matched ECC sample, pooled across all years from 2001-2021 and restricted to working age adults. The median annual earnings in this sample is \$16,939.88 and the modal primary sector of employment is "Temporary Help Services" (NAICS code 561320).

²³This data covers the vast majority of payroll employment, but there are a few entities (e.g, some churches) that are exempt from unemployment insurance. There are other limitations to wage records in Connecticut that are common to many state administrative data systems: DOL records do not contain W-2 income from working for employers in other states, wage records for federal employees, or wage records for individuals employed in the informal or gig economy.

²⁴An alternative explanation is that the individual's state of residence is not Connecticut or that the individual is either not yet born or deceased.

Figure C.3: Conditional earnings density among ECC applicants

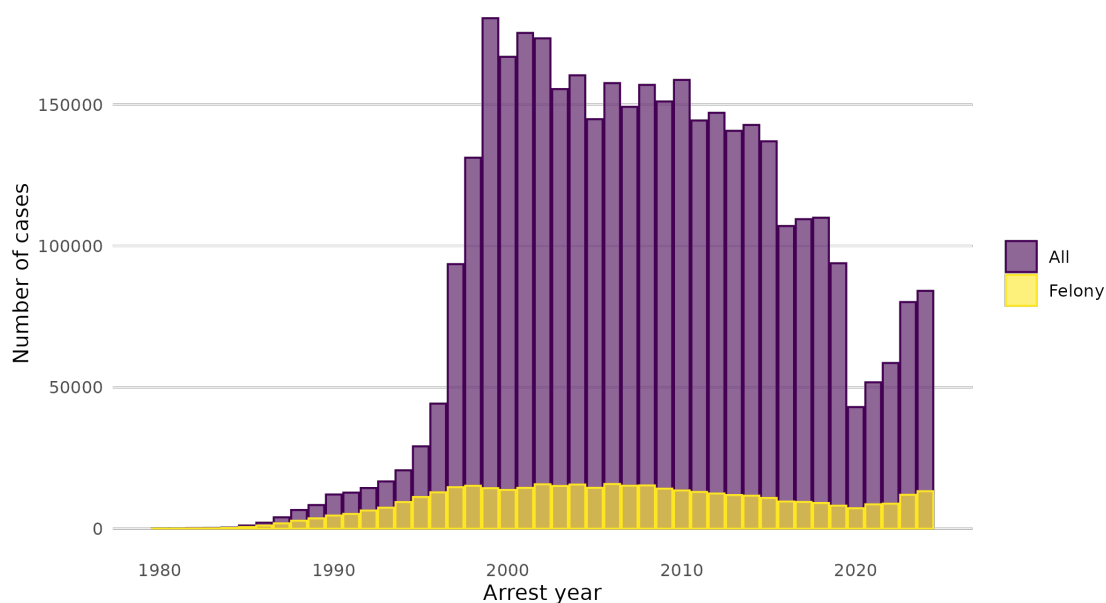


Note: The x-axis is individual annual earnings, reported in 2023 dollars and shown on a log scale. The density is at the individual-by-year level for all years from 2001-2021. The dashed yellow line reports the median earnings conditional on employment, which is \$16,939.88. The data source is the sample of ECC applicants that are uniquely matched to DOL earnings records, restricted to individuals ages 18 to 65 in a given year. The unemployment rate (i.e., the share of individual-year combinations with zero earnings) is 58.2%.

C.3 Criminal court data

In this section, I describe the criminal case records provided by the State of Connecticut Judicial Branch (CJB). I use the September 9, 2024 daily refresh of cases from CJB containing 3,778,246 criminal and motor-vehicle infractions. These records contain pending cases and conviction records for criminal and motor-vehicle matters, including felonies, misdemeanors, infractions, and violations. The data include case-level, defendant-level, charge-level, and sentencing files, which I link using docket identifiers, but does exclude sealed, erased, pardoned, clean slate, and otherwise non-disclosable cases. Historical coverage is limited for cases disposed before 2000, which were often archived after sentence completion and may not appear in the current refresh. Because my sample begins in 2003, this does not affect data availability in this study. Figure C.4 shows the distribution of convictions by arrest year.

Figure C.4: Connecticut criminal and motor vehicle infractions



Note: The y-axis is the count of convictions involving criminal or motor vehicle infractions in Connecticut. Separate series are included for all cases and cases involving at least one felony charge. The x-axis is the calendar time in years from January 1980 to 2024. The data source is the full sample of pending cases and conviction records available in the State of Connecticut Judicial Branch's daily criminal database refresh. Sealed, erased, pardoned, archived, and otherwise non-disclosable cases are not included. Cases disposed prior to 2000 are incompletely observed due to archival rules.

D Measurement appendix

D.1 KEI measurement model

Let $Y_i = (Y_{iN}, Y_{iL}, Y_{iG}, Y_{iC}, Y_{iP}, Y_{iS})'$ collect the standardized numeracy, literacy, language, creativity, physical, and personal/social scores for child i . I assume that these six scores represent noisy measures of two underlying factors such that

$$Y_i = \nu + \Lambda\eta_i + \varepsilon_i, \quad \eta_i = \begin{pmatrix} \eta_i^A \\ \eta_i^D \end{pmatrix}, \quad (6)$$

where η_i^A is an academic readiness and η_i^D is developmental readiness. The (assumed) factor loading matrix is

$$\Lambda = \begin{pmatrix} \lambda_{NA} & 0 \\ \lambda_{LA} & 0 \\ \lambda_{GA} & 0 \\ 0 & \lambda_{CD} \\ 0 & \lambda_{PD} \\ 0 & \lambda_{SD} \end{pmatrix}. \quad (7)$$

In other words, numeracy, literacy, and language load only on the academic factor, while creativity, physical skills, and personal/social skills load only on the developmental factor. While each score loads onto only one factor, the two factors may be correlated:

$$E[\eta_i] = 0, \quad \text{Var}(\eta_i) = \Phi = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}. \quad (8)$$

I assume local independence of the residuals, so $\text{Var}(\varepsilon_i) = \Theta$ is diagonal and $\text{Cov}(\eta_i, \varepsilon_i) = 0$. I also fix both factor variances to one to measure estimated effects in standard deviations of the latent factors. To interpret the factors, I normalize the sign on the academic (developmental) factor so the loading on numeracy (creativity) is positive. I use Gaussian maximum likelihood to estimate the six factor loadings, the factor correlation, and the six variances of the measurement model residuals.

To recover estimates of the effects of assistance on these latent factors, I then estimate the instrument's reduced form effect on each score jointly, retaining the household-clustered covariance matrix across the six estimates. Finally, I project the score-level reduced forms onto the estimated factor-loading space and divide the resulting latent factor reduced forms by the common first stage.